



Development of a Framework for Analyzing Corporate Financial Structured and Unstructured Data Using Artificial Intelligence

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Abstract

Corporate annual reports are a combination of structured financial data (income, balance sheet, etc.) and large amounts of unstructured text information in a single document. While the financial data is structured, it is also embedded within the unstructured parts of the report, and a system that could understand both structured and unstructured financial data at the same time would be required for a clean separation of these elements (Duane et al., 2025; Li et al., 2025). This structured financial information exists in an embedded structure and has never been possible to extract this structured financial information and present it in a structured and comparable output format using a completely on premise deployed, conversational AI that has active hallucination mitigation capabilities.

To solve this problem, this research proposes a framework based on retrieval augmented generation and large language models, which consists of 5 layers: an input/output layer, a chatbot based RAG layer, an agent layer, a validator layer, and an analytical system. The framework sees an annual report as one single hybrid source that has to be processed as a whole, both the financial information is structured and the textual narrative containing the material is unstructured, and the LLM strictly operates on the structured data at all times, thereby mitigating the risk of hallucination in the output numbers. The research methodology used in the research is design science research, which has literature review, CRISP DM, and expert interviews based on the TOE framework.

The artifact demonstration showed that all ten functional requirements were met and that the artifact was evaluated as having a 7 out of 10 performance in the verification assessment. Five domain practitioners were interviewed as experts and confirmed the technical integrity of the architecture, the utility of the conversational interface for non technical users and the need for on premise deployment model in regulated corporate environments.

Table of Contents

List of Abbreviations.....	5
List of Figures	6
List of Tables	7
1 Introduction	1
1.1 Motivation.....	1
1.2 Problem Definition.....	2
1.3 Objective	3
1.4 Research procedure.....	3
1.5 Structure of the work	3
2 Overview on Artificial Intelligence.....	4
2.1 Definition of Artificial Intelligence.....	4
2.2 Applications of Artificial Intelligence	5
2.2.1 Artificial Intelligence in Business and Finance	5
2.2.2 Artificial Intelligence in Data Management.....	6
2.3 Fields of Artificial Intelligence	7
2.3.1 Machine learning	7
2.3.2 Large Language Models	8
2.3.3 Retrieval Augmented Generation	9
3 Corporate Financial Data Analysis	10
3.1 Definition of corporate Financial Data	10
3.2 Structured Financial Data.....	11
3.3 Unstructured Financial Data.....	12
3.3.1 Types and sources	12
3.3.2 Challenges of Processing Unstructured Data	13
4 Literature Review	14
4.1 State of the Art	14
4.2 Comparative Analysis	24
4.3 Scientific Gap	28
4.4 Research Question	29
5 Methodology Fundamentals	30
5.1 Basic Scientific Methodology.....	30
5.1.1 Design Science Research Methodology	31
5.1.1.1 Offerman Design Science Approach	31

5.1.1.2	Hevner Design Science Approach	32
5.1.1.3	Peppers Design Science Approach	34
5.1.2	Qualitative Interviews.....	35
5.1.2.1	Delphi Method.....	36
5.1.2.2	Technology Organization Environment Framework	37
5.1.3	Literature Review.....	38
5.1.4	Data Science Research Methodology	38
5.1.4.1	Cross Industry Standard Process for Data Mining Framework	38
6	Solution Approach.....	40
6.1	Problem Identification.....	42
6.1.1	Identifying the Problem.....	42
6.1.2	Literature Review.....	42
6.1.3	Comparative Analysis.....	43
6.2	Solution Objective	43
6.3	Solution Design	44
6.3.1	Analysis	45
6.3.1.1	Functional Requirements	47
6.3.1.2	Non Functional Requirements.....	49
6.3.1.3	Partial solution	50
6.3.2	Conceptual Design	51
6.3.2.1	Behavioral Perspective	52
6.3.2.2	Structural Perspective	55
6.4	Evaluation	58
6.4.1	Artifact Demonstration	59
6.4.1.1	Validation	68
6.4.1.2	Verification	69
6.4.2	Experts Interviews	70
7	Limitation and Future Recommendations.....	74
7.1	Limitations related to the concept.....	74
7.2	Future Recommendations	75
8	Conclusion	76
	References.....	78
	Appendix A.....	84

List of Abbreviations

ACFRs	Annual Comprehensive Financial Reports
AI	Artificial Intelligence
BPMN	Business Process Model Notation
CRISP DM	Cross Industry Standard Process for Data Mining
CSV	Comma Separated Values
DSRM	Design Science Research Methodology
ERP	Enterprise Resource Planning
LLM	Large Language Model
NLP	Natural Language Processing
PDF	Portable Document Format
RAG	Retrieval Augmented Generation
SQL	Structured Query Language
TOE	Technology, Organization, and Environment

List of Figures

<i>Figure 1. Proposed Research Process by Offermann et al., 2009.....</i>	<i>32</i>
<i>Figure 2. Proposed Research Process by Hevner, 2004Error! Bookmark not defined.</i>	
<i>Figure 3. Proposed Research Process by Peffers, 2007.....</i>	<i>35</i>
<i>Figure 4. TOE Framework adopted from Tornatzky et al., 1990.</i>	<i>37</i>
<i>Figure 5. Proposed Research Process by Kannengiesser & Gero, 2023.....</i>	<i>39</i>
<i>Figure 6. Solution approach adapted from Offermann et al. (2009), Hevner (2004).....</i>	<i>40</i>
<i>Figure 7. BPMN Workflow of Chatbot Based RAG for Analyzing Corporate Financial Data....</i>	<i>53</i>
<i>Figure 8. Sequence Diagram of Chatbot Based RAG for Analyzing Corporate Financial Data.</i>	<i>54</i>
<i>Figure 9. User Activity Diagram</i>	<i>55</i>
<i>Figure 10. Conceptual Model of Chatbot Based RAG for Analyzing Corporate Financial Data</i>	<i>57</i>

List of Tables

Table 1. Comparative Analysis..... 27

1 Introduction

Artificial Intelligence is one of the possibly disruptive factors in business finance analysis. Corporate annual reports are a hybrid type of data, which include both structured financial data – such as income statements and balance sheets – and large volumes of unstructured text in the same document (Duane et al., 2025; Li et al., 2025). The structured financial information is embedded in these reports, but without a system that can process structured and unstructured information at the same time, it is a source of information that is largely untouched because of technical difficulties in processing structured information in the context of unstructured information (Shilpa and Shambhavi, 2022). Large language models (LLMs) and retrieval augmented generation systems have shown great promise in handling unstructured text and extracting significant information from financial documents (Kong et al., 2024; Srinivasan et al., 2025). Current AI technologies are either qualitative analysis of the textual content or useful only after the financial data has been processed, not to solve the problem of reliably extracting financial data from within mixed format documents (Yang, 2022; Al et al., 2025). The fact that LLMs can produce plausible but incorrect information with a high degree of certainty, which can lead to hallucinations, also exacerbates the issue, as does the demand for data sovereignty in many corporate settings which renders cloud based solutions unworkable (Kong et al., 2024; Al et al., 2025).

1.1 Motivation

The motivation for this paper is based on the understanding that business financial data analysis is largely based on extraction of financial data from annual reports manually. Traditional manual methods are time-consuming, prone to mistakes, and unable to handle today's demands (Yang, 2022). This realization that corporate annual reports are composed of structured financial information and unstructured textual narratives has created the need for such systems that can deal with the two forms as a single combined hybrid source, rather than models for each form alone. (Duane et al., 2025). While the potential of LLMs and RAG to address the challenges of financial data processing is promising, there exists a critical gap between the technology's capabilities and its ability to be practically implemented, especially when data sovereignty and explainability are part of the mandate. The promise of LLMs and RAG in overcoming the hurdles of financial data processing has been significant, however, the technological solutions and their practical application is still quite distant, particularly when data sovereignty and explainability are in demand.

1.2 Problem Definition

Even though computers have gotten better at understanding language and can help with analysis a big problem still exists. There is no system that can take information from company annual reports, which are partly structured and partly written in plain text and present it in a clear and comparable format through a conversational AI interface. This system needs to be installed on the company's servers with features to prevent the AI from making up information.

Company annual reports are a mix of financial data like balance sheets and income statements and unstructured text. This makes it hard for computers to process them automatically because the financial data can't be easily separated from the rest of the text. If the AI makes a mistake with financial numbers it can lead to bad decisions and big financial losses.

Some approaches try to improve the accuracy of AI responses. They focus on general answers and don't help extract specific financial data from mixed-format documents. The problem gets worse because of data protection rules, which prevent financial data from being sent to external servers. Existing solutions that can be installed on premises don't help extract structured data from mixed-format documents.

1.3 Objective

The main objective of this research is to create a framework that uses AI and language models to extract financial data from mixed-format annual reports and turn it into a clear and comparable format. This framework should allow for fact-based questioning, all within an on-premise environment. The framework addresses the issue that AI models are not reliable for direct numerical data extraction due, to the risk of hallucinations.

1.4 Research procedure

The paper designs a framework The framework separates the tasks of extracting, validating and interacting with the data ensuring the AI works with verified structured data. The research paper uses design science research methodology to highlight the challenges of extracting financial data from mixed-format corporate annual reports. The research starts with a literature review to identify a gap and confirm the research question. then a framework design is created, followed by building and evaluating an artifact to confirm that the solution covers the research question and objectives.

Structure of the work

The paper is organized as follows. [Chapter 2](#) provides an overview of artificial intelligence, including its definition, applications in business, finance, and data management, and its major fields such as machine learning, large language models, and retrieval augmented generation. [Chapter 3](#) focuses on corporate financial data analysis, explaining the definition of corporate financial data, the distinction between structured and unstructured data, the types and sources of unstructured financial data, and the challenges of processing unstructured information. [Chapter 4](#) presents the literature review, outlining the state of the art through five selected studies from 2022 to 2025, followed by a comparative analysis that identifies the scientific gap driving this

research. [Chapter 5](#) introduces the methodology fundamentals, covering design science research methodology through Offermann et al. (2009), Hevner (2004), and Peffers et al. (2007), alongside qualitative interview methods and CRISP DM. [Chapter 6](#) presents the solution approach through problem identification, solution design, and evaluation comprising artifact demonstration and expert interviews. [Chapter 7](#) addresses the limitations of the proposed framework and future recommendations, and [Chapter 8](#) concludes the research. The paper closes with references and appendices as required.

2 Overview on Artificial Intelligence

There are various definitions for the term “Artificial Intelligence”. This chapter is going to address divergent definitions that have been discussed in previous articles and literature reviews. In addition to exploring its applications and different fields more closely.

2.1 Definition of Artificial Intelligence

The concept of Artificial Intelligence (AI) does not have any unified, universally accepted definition at the academic, governmental, and community levels, which can contribute to inconsistencies in the perception of the technology and its study (Kelly et al., 2023:2). Lots of explanations refer to AI as being the machines working on the tasks that would otherwise be handled by human intelligence, including visual perception, speech recognition, and problem solving (Cox and Mazumdar, 2024:2). Nevertheless, this analogy to human power is abstract, and it does not consider AI as a technology but as a developing concept to which cultural meanings are attributed to it (Cox and Mazumdar, 2024:3).

AI is most appropriate to consider to imply a package of technologies, and machine learning is one of the most significant elements of this definition (Cox and Mazumdar, 2024:3; Chiu et al., 2024:1). It is also called a machine based system capable of, based on the predetermined human based goals, predicting, offering suggestions, or judgements based on the data (Chiu et al., 2024:5; Kelly et al., 2023:2). This can be described as employing algorithms to find the pattern in massive volumes of information that allow the system to enhance itself as time passes without direct instructions on the exact tasks to perform (Yang, 2022:2, 5). Existing implementations are currently classified as Artificial Narrow Intelligence (ANI), meaning task dependent, and the future Artificial General Intelligence (AGI) will be able to transfer learning across different situations (Cox and Mazumdar, 2024:2).

2.2 Applications of Artificial Intelligence

This section will explore how artificial intelligence is being applied in practice across two key areas: business and finance, and data management. Exploring these drivers will help to get better understanding of how Artificial Intelligence affect today's world.

2.2.1 Artificial Intelligence in Business and Finance

AI in business has completely revolutionized the strategic decision making in different businesses. Using AI technologies, especially machine learning algorithms and prediction models, organizations can handle large amounts of data and determine intricate patterns that organizations cannot recognize based on traditional analytical tools (Vudugula et al., 2023, 2). The capabilities enable the business to shift their decision making mode toward proactive decision making rather than reactive using real time and historical data to predict the outcomes and the optimal resource allocation. The artificial intelligence approach like Random Forest and Gradient Boosting has also proven to be quite effective in segmentation of customers and predicting demand, enabling companies to improve their competitive place by using information (Fallahi et al., 2023, 5). Moreover, the models of deep learning such as Long Short Term Memory networks have also turned out to be useful when there is a complex pattern and sequence in the data, which causes the enhancement of the predictability of the strategies and lessening the operational risks (Vudugula et al., 2023, 11).

Artificial intelligence in the financial sector has been a game changer that has changed the way in which institutions relate with customers and evaluate risks. Such AI models as ChatGPT have transformed customer interactions by offering personalized financial recommendations and giving real time answers to their queries using AI based chatbots (Rane, 2023, 2). The systems use natural language processing systems to obtain answers to complex financial inquiries and increase customer satisfaction, at the same time minimizing operational costs. Moreover, AI based predictive models have become essential in fraud detection as they constantly check transactional data to detect any pattern that hints at fraudulent activity to help institutions to proceed with strict protective measures (Rane, 2023, 3). Application of information fusion technology in detection of e commerce fraud has demonstrated that the identification accuracy is much higher because these systems are utilised to process information about financial risks consisting of more than one sources at the same time, which makes their assessment of the potential fraudulent activities more reliable (Li, 2022, 7).

2.2.2 Artificial Intelligence in Data Management

The rise of Artificial Intelligence (AI) in data management systems has fundamentally changed the process and protection of information by organizations. One interesting example is how to optimize data pipelines to maximize efficiency and privacy, by citing the Artificial Intelligence based Data Management System (AI DMS) of smart cities. The concept uses the Principal Component Analysis (PCA) to process raw data by feature mapping, selecting, normalising, and converting these data to ensure that sensitive data is safeguarded prior to data publication (Jyothi et al., 2024:7). The system also introduces a multi level sensitivity model which categorizes data as sensitive, quasi sensitive, and public, allowing one to share data more delicately besides having a solid security level in place (Jyothi et al., 2024:5).

In addition to the preservation of privacy, AI can dramatically improve the data integrity with the help of sophisticated analytical approaches that are a part of Master Data Management (MDM). Machine learning algorithms are automatic, which means that traditionally manual processes, including data matching and detecting anomalies, can be automated, eliminating the scalability constraints of rule based application (Tadi, 2020:11). These artificial intelligence powered features allow activating proactive data management by detecting issues that might lead to adverse experiences earlier than possible, whereas natural language processing makes it easier to incorporate unstructured data into consistent management systems (Chen et al., 2023:7).

2.3 Fields of Artificial Intelligence

This section will focus on three core fields of artificial intelligence that are most relevant to this research: machine learning, large language models, and retrieval augmented generation. We will further explore how each of these fields contributes to the analysis of financial data and why understanding them is essential to the approach proposed in this work.

2.3.1 Machine learning

Machine learning, one of the most popular sub domains of artificial intelligence, is inherently the development of computational methods that allow computer systems to learn automatically without requiring a program to be written that instructs them on each task they perform. According to Pandey et al. (2022:1) the process is to make a system intelligent using the previous experiences so that the system can complete its decision making tasks independently. Machine learning is conceptually a process which is divided into two steps: training a model with a dataset and testing or choosing a decision on new unseen data with the help of the trained model (Pandey et al., 2022:1). This style is a transition to traditional programming, where a system does not write instructions, but directly gathers patterns and rules based on the examples provided to it (Nasteski, 2017:3). Thus, the key concept of machine learning is that it is adaptive in nature, meaning that models are constructed by applying algorithms to known data in order to derive an insight into unknown data (Kuhl et al., 2022:4).

There are various machine learning approaches that are broadly defined through the use of three primary learning signals to broadly organize them. The most widespread is supervised learning, which trains algorithms on labeled data, that is, the desired output is known, to make predictions on new data, which makes it appropriate in classification and regression tasks (Pandey et al., 2022:1-2). On the contrary, unsupervised learning is concerned with unlabeled data, where the algorithm is asked to make implicit assumptions about the underlying structures or groupings, like putting similar objects into a single cluster and no external instructions are provided (Wei et al., 2019:346). Lastly, the reinforcement based learning involves a dynamic trial and error mechanism by which an agent learns to make decisions through exposure to an environment and reinforcing its accumulating performances overtime (Pandey et al., 2022:6). These different paradigms enable the machine learning to be used in a wide extent of issues, both in classification and planning.

2.3.2 Large Language Models

Large Language Models (LLMs) are advanced AI systems designed to understand and generate human language by processing huge amounts of text data. In contrast to the previous natural language processing models that failed to achieve basic understanding, modern LLMs such as GPT-4 can discuss, respond to queries, and write texts that sound resembling human work to an unparalleled extent (Surden, 2024:1942). Such models are designed on the basis of the transformer architecture that employs self-attention mechanisms that enable them to analyse the contexts related to what the user has input and identify the words that are the most relevant and thus help it to produce coherent responses (Surden, 2024:1963-1964). The most striking feature of LLMs is that, essentially, they are next word prediction models, but because of their ability to scale to billions of parameters, they have generated emergent behaviors such as reasoning and problem solving that were not modeled (Makridakis et al., 2023:537).

LLM training requires two important stages that facilitate their advanced outputs. The first is pretraining, the model is trained on vast amounts of text, such as Wikipedia, books and the internet, learning statistical patterns of language by repeatedly attempting to predict missing words and update its own internal parameters (Bonner et al., 2023:30 31). This self-supervised method is useful because it lets the model build knowledge without the need of manually labeled data. Second, approaches such as instruction fine tuning and reinforcement learning based on human feedback can be used to convert these general language models into systems that can obey arbitrary instructions and generate responses that correspond to the human preferences (Surden, 2024:1967 1968). Despite these capabilities, LLMs are also associated with key weaknesses, such as susceptibility to "hallucinations," for example they create a convincing, yet factually incorrect, information, making them unreliable as an immediate replacement of search engines (Makridakis et al., 2023:538).

2.3.3 Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) is a new category of hybrid frameworks which are meant to overcome the core constraints of generative artificial intelligence, namely the issue of large language models generating hallucinated responses that are not rooted in the real world (Bansal and Suddala, 2024:1). RAG systems also readily consult external sources of knowledge, for example document databases or web search engines, at inference time, in order to access pertinent information that conditions the generation process, unlike conventional models of autoregressive models that only use their internal parameters to generate text (Genesis, 2025:2 3). This is a reliable retrieval mechanism that the generated content is based on the attributable and up to date information, and factual accuracy and contextual relevance have greatly risen (Zhao et al., 2026:2). This core architecture works in two steps, the first step is to locate useful passages in a knowledge base by a retriever based on various methods such as dense passage retrieval and the second step is to generate consistent outputs by a generator that works with the identified passages and the initial query (Bansal and Suddala, 2024:4 5).

The combination of retrieval and generation takes place in ways that are methodologically diverse and which define the way external information complements the performance of the model. The most common paradigm, query based RAG, has a smooth interface with the retrieved content and the original query of the user being inputted at the retrieval stage which is directly fed into the generator as a composite prompt (Zhao et al., 2026:6). Other frameworks can be latent representation based RAG, which introduces retrieved objects as hidden representations via other processes such as cross attention and logit based RAG, where the probability of retrieved distances are combined in the process of decoding (Gargari & Habibi, 2025:2). Such varied research augmentation strategies allow the RAG systems to retain the of generative models and capitalize on the accuracy of compensatory information retrieval so that the discontinuity existing between parametric knowledge captured in model weights and various dynamic and external sources of information is overcome (Genesis, 2025:5 6).

3 Corporate Financial Data Analysis

This chapter will address the definition of corporate financial data and examine its two main forms: structured and unstructured data. It will further explore how artificial intelligence can be applied to analyze each data type and how combining both forms offer a more comprehensive approach to corporate financial analysis.

3.1 Definition of corporate Financial Data

Corporate financial data may be widely conceptualized as the quantitative data, which represents the economic operations of a company, and represents the basis of analysis and decision making. The information is mostly based on the corporate financial statements that give a systematic account of the financial status and performance of a firm (Krylov, 2023:5). Such data comprises conventional data such as revenue, profits, and assets value, which typically help to compute the most important financial ratios to evaluate liquidity, profitability, and solvency (Krylov, 2023:5 6). The capabilities of this data, however, have increased dramatically in the new age where it now includes real time records of transactions, market data, and information provided by various sources such as social media that add to the dynamic vision of the financial health of a company (Cai and Liu, 2024:2).

Moreover, the nature of corporate financial data is essentially linked to the concept of financial disclosure, which involves the deliberate release of this information to outside parties. This disclosure may be in numbers or also in qualitative form and is disseminated with the help of different formal and informal means, including annual reports, press releases, and conference calls (Hassan and Marston, 2019:2). The data itself is never just a sequence of numbers as it is a complex structure that is measured in different dimensions, such as its quantity, quality, and even tone which researchers strive to measure in diverse ways (Hassan and Marston, 2019:8 9). Finally, the information becomes the main source of communication between management and investors, and therefore accurate representation and analysis of it become crucial to the market operation (Hassan and Marston, 2019:2).

3.2 Structured Financial Data

Structured financial data is that data which is organized in a predefined format that can easily be accessed in order to process it and conduct quantitative analysis on it. According to Whitelock (2018:82), structured data involves data stored in the customer relationship management systems, enterprise resource planning systems, legacy databases, as well as the third party applications, and can be measured on dimensions such as volume, velocity, variety, veracity, validity, volatility, and value to ascertain its applicability in decision making processes. Such data is especially useful in a financial analysis since it is in standardized formats allowing it to be stored, recalled and analyzed using known analytical tools (Whitelock, 2018:82).

Essentially, structured financial data is applied practically in organizational contexts, where there is the use of financial statements and transaction records to evaluate activities within the organization and identify abnormalities. As shown by Whitelock (2018:89), there are structured financial statement data, which throughput, inventory, operating expense, sales, net income, return on investment, and cash flows are measurement metrics, which can be systematically analyzed in order to evaluate the change in operational and financial performance over time. Tan et al. (2025:2) further go on to explain that financial transaction data is usually made of a variety of structured elements arranged in tabular form, comprising of both numerical characteristics such as transaction amounts, account balances and credit limits, as well as categorical characteristics such as client data and geographic locations and transaction types which make up the basis of applications of fraud detection and risk assessment.

3.3 Unstructured Financial Data

This section will explore the types and sources of unstructured financial data found in corporate environments. We will further examine the key challenges involved in processing this type of data and how understanding these challenges is essential to justifying the use of advanced AI techniques in corporate financial analysis.

3.3.1 Types and sources

The widespread development of unstructured financial data poses serious problems to the stakeholders due to what it is and where it gets its source. Unstructured data is widely defined by the absence of a specific format, usually presented in the form of the text based content that can be hard to operate with the conventional tools (Shilpa and Shambhavi, 2022:2). The most common structure of such data is the Portable Document format (which is often referred to as PDF), through which fiscal indicators are regularly reported in government Annual Comprehensive Financial Reports (ACFRs) and emission levels are reported in corporate environmental, social, and governance (ESG) reports (Li et al., 2025:2). In addition to these, the less regular textual types include social media discussions, open ended surveys, and news blogs, which also hold opinions, the discussed topic, and recommendations of value to analyze but difficult to harvest in bulk (Shilpa and Shambhavi, 2022:2). Moreover, the financial domain is full of documents, such as contracts, where essential data, such as prices and negotiated volumes, is organized, and analyst reports, which contain qualitative estimates, not an element of standardized data sets (Li et al., 2025:2).

This unstructured data also has diverse origin sources, and it makes the financial analysis and research more complicated. News and social media materials produce enormous volumes of textual information on market feeling and real time happenings and are becoming utilized to perform a variety of tasks such as forecasting price fluctuations in the stock market (Shilpa & Shambhavi, 2022:3). Formally, regulatory filings like the risk factor disclosure portions of corporate quarterly and annual reports are vital sources that include discussions of potential business and financial risk that have been proven to drive stock returns (Cong et al., 2024:47). Financial sector, also, deals with data in the area of specialization, which is why, in spite of their intent to be approached in a structured manner, most of the time it has to be paid out of the non machine readable format (Li et al., 2025:9).

3.3.2 Challenges of Processing Unstructured Data

The use of process mining on unstructured financial data, including transactional emails, scanned invoices or recordings of days spent in a call center introduces major initial challenges which center principally on data quality and meaning. One of the main challenges is that noisy and incomplete data is common and has flaws such as a mistake in recordings or the lack of a certain value that can result in inaccurate calculations (Adnan and Akbar, 2019:13). Moreover, this information is not in a prescribed format and thus, it is hard to structure and analyze as compared to ledger based structured data (Badgujar, 2020:1). This raw data should be converted according to the requirement to be usable in the process of discovery to assign a distinguishing case number to the individual financial transaction, which is anything but easily achievable once the data is generated by the different sources (Kamm et al., 2021:4).

The key problem with such divergent formats lies after the preliminary preparation of the data and the retrieval of meaningful and contextual information. The diversity of financial data which may include text, images and audio is a challenge as it is hard to extrapolate and compare data to single perception of a financial process (Adnan & Akbar, 2019:14). As an example, although an image of an invoice could be subject to a tremendous amount of information being extracted, a large divide would need to be transcended between the raw visual features and the activities in the realm of business to comprehend the semantic meaning of the image and place it appropriately in the context of a particular payment procedure (Adnan and Akbar, 2019:10; Kamm et al., 2021:3). This is further complicated by the fact that it must encompass domain related financial knowledge so as to be able to properly interpret events to come up with the extracted insights being correct and reliable to enable appropriate decision making.

4 Literature Review

This chapter aims to identify a scientific gap in the field of "analyzing corporate financial structured and unstructured data using artificial intelligence." To achieve this objective, a literature review is conducted as the primary scientific method. The process is divided into two steps. First, the state of the art is established by presenting recent research and studies related to the topic. Second, a discussion is carried out in the form of a comparative analysis of the presented works, which serves to extract the scientific ga

4.1 State of the Art

The primary objective of this section is to provide an overview of the most current and relevant academic papers on the topic of **“Analyzing corporate financial structured and unstructured data using artificial intelligence”**. The selection process prioritizes papers that investigate the technological implementation of artificial intelligence models for the analysis of both structured and unstructured financial data. The literature search was conducted using the following keywords: **“Artificial Intelligence”, “Finance”, “Data analysis”, “Structured data”, “Unstructured data”, “Machine Learning”, “Retrieval Augmented Generation”, and “Large Language Models”**. The search was limited to articles published between **2022 and 2025** and encompasses empirical studies, technical model based research, conceptual analyses, bibliometric reviews, and literature reviews. An initial search using Google Scholar and the Google search engine yielded a total of **21 articles**. From these, five papers were selected based on their relevance, originality, and the absence of duplicates.

Large Language Models for Financial and Investment Management: Models, Opportunities, and Challenges (Kong et al., 2024)

The **motivation** of this article is to understand how the adoption of LLMs in financial and investment management representing a main shift in how the industry processes information. Unlike traditional quantitative models, LLMs can process unstructured data like subtle contextual nuances and generate insights that blend computational analysis with qualitative understanding, a capability particularly valuable in an industry where interpreting complex information streams is critical.

The **research problem** addressed is the intersection of significant opportunities and challenges that come from applying LLMs in finance. Key issues include signal decay, where the effectiveness of LLM generated strategies may decrease as more market participants adopt them. data pollution from inaccurate or LLM generated content and the complexity of managing high dimensional financial data. The issues of modeling include the speed of inference and cost, use of look ahead bias in back testing, and the possibility of hallucinations. Moreover, such ethical aspects as benign alignment, legal responsibility, data privacy, and revelation of incentives cannot be ignored.

The **aim** of this survey is to provide a comprehensive overview of the current state of LLMs in financial and investment management, examining various models developed specifically for financial applications, their architectures, capabilities, and limitations. The authors seek to explore why LLMs are particularly well suited for financial tasks and to critically assess both the challenges and opportunities at their intersection.

The **scientific method** employed is a thorough literature review of financially specialized LLMs, categorized by their underlying architectures. The paper discusses the GPT series models and frameworks which include Ploutos, which integration of multimodal data and the development of explainable rationales when making predictions about stocks. It evaluates models based on BERT such as FinBERT that do sentiment analysis using deep bidirectional models; T5 based models such as BBT FinT5 language understanding and generation; ELECTRA based models such as FLANG combined with a discriminator generator system;

BLOom based models such as BloombergGPT; and ELECTRA based models such as FLANG and InvestLM. Other adaptation methods discussed in the review are fine tuning, instruction tuning, zero shot learning, and low rank adaptation (LoRA).

The **findings** indicate that financial domain specific LLMs offer significant benefits for financial analysis, including enhanced contextual understanding of financial terminology, transfer learning flexibility that reduces reliance on massive domain specific datasets, and scalability for real time analysis. Multimodal capabilities can be used to blend various data categories which include text and charts. Other models like FinBERT have become a new standard in the field of financial text analysis, and other frameworks like Ploutos have shown to be better predictors with interpretable reasons why. There are more affordable and easier to access solutions offered by open source models, such as FinGPT, than a proprietary model.

The **limitations** identified include potential expert selection bias in complex models, computational complexity, and questionable performance on high dimensional financial time series data. There is also the challenge of evaluating LLM performance in markets fundamentally altered by the technology itself, as current benchmarks predate the widespread adoption of LLMs. Domain specific models might be less effective on general NLP tasks, and the lack of interpretability remains a significant concern. Additionally, LLMs are susceptible to producing hallucinated or factually incorrect statements.

Future research directions include developing adaptive LLMs capable of continuously updating to new data and market conditions to mitigate signal decay, creating new and versatile benchmarks that reflect current market realities, and enhancing data quality assessment procedures for LLM generated content. The authors underline the necessity of cooperation among academic, industrial, and regulatory circles to develop the technology without violating the ethics and law.

A Review of Financial Data Analysis Techniques for Unstructured Data in the Deep Learning Era: Methods, Challenges, and Applications

(Duane et al., 2025)

The **motivation** for this review is the explosion of unstructured financial data from sources such as news articles, social media, corporate filings, earnings call audio, and satellite images. Traditional quantitative models that rely on structured numerical data struggle to process these complex information sources effectively.

The **research problem** addresses the challenges of adopting deep learning in finance. The financial regulators require explainable and clear models, but the deep learning models usually use black boxes. In finance labeled data are both rare and expensive to acquire. Data privacy and compliance issues further complicate the use of unstructured data. The review fills a gap in existing literature by providing a focused synthesis on deep learning's impact on unstructured financial data across multiple modalities.

The **aim** is to provide a comprehensive review of deep learning techniques for unstructured financial data. authors structured the discussion by the data modality like textual, audio and visual data and technique including natural language processing models, speech processing, multi modal fusion and transformer architecture across applications such as sentiment analysis, fraud detection and credit risk assessment.

The **scientific method** employs a data modality based literature review. In the case of textual information, the review analyzes RNNs, LSTMs, and CNNs and transformer based models such as BERT and its financial version FinBERT. In the case with audio data, it addresses speech recognition models, as well as audio transformers like Wav2Vec 2.0. In the case of visual data, it talks about 2 D CNNs and vision transformers of satellite images and documents. It also makes a review of generative models such as GANs and VAEs.

The **findings** demonstrate that deep learning models achieve superior performance in credit risk prediction and fraud detection compared to earlier approaches. FinBERT performs much better on financial sentiment analysis as compared to general domain BERT. Multimodal fusion of text and numerical data posted a score of 0.91 with numeric data, and 0.808 as a single form of data. Social media sentiment integration enhances returns and volatility forecasts, while unstructured data analysis improves fraud detection and trading performance.

The **limitations** include scarce high quality labeled data, ambiguous and noisy labels, and black box models that challenge interpretability and regulatory compliance. Privacy of data inhibits the use of customer communication training. The danger of unfair discrimination exists with the help of protected attribute proxies. Financial data is not stationary and therefore the models fail when in a regime change. Big models are expensive to compute and their implementation on premise is costly, and any integration with legacy systems is technically complex.

Future research directions include developing finance oriented foundation models and multimodal models that jointly understand text, time series, graphs, and images. There is a need to enhance interpretability with the help of modular neural networks and causal inference. It is suggested to construct hybrid systems that involve deep learning extraction and conventional financial approaches. The news about related parties can support the evaluation of the entire corporate networks using Graph Neural Networks with text encoders.

Designing for Explainability and Data Sovereignty: A Design Principles Approach for LLM Augmented FinTech Analytics (Al et al., 2025)

The **motivation** is the tension in FinTech between the push for data driven decision making and a persistent analytical skill gap. The non technical stakeholders are not trained to utilize the complicated predictive tools, and thus the potential value is not fulfilled. Simultaneously, financial institutions have to deal with tight control of regulations and data sovereignty requirements that define the way in which data may be stored and processed.

The **research problem** is that existing analytical tools lack transparency, causing hesitation in regulated settings where justifying decisions is a legal requirement LLMs available on clouds have the potential to assist users who are not technical experts, however, they necessitate them to submit sensitive financial information to external APIs, which is prohibited by the principles of data sovereignty. It is impossible to have an artifact with great predictive accuracy, explainability to non technical users, and on premise data sovereignty.

The **aim** is to design and develop the Scalable LLM Augmented Data Analytics Framework (SDAF), a system that balances advanced analytics with governance requirements. SDAF uses locally hosted LLMs such as LLaMA or Qwen to provide natural language guidance, explaining what models do and why results appear, while keeping all data within the organization's perimeter.

The **scientific method** uses Design Science Research Methodology (DSRM) with an Agile development process. It was done on an iterative prototyping basis with user feedback. Assessment and technical standards with FinTech style data were used to assess the architecture against its requirements by conducting scenario based walkthrough tests. The findings show that SDAF can address its six Design Requirements.

The **findings** show that SDAF can address its six Design Requirements. Technical benchmarks demonstrated that the modular architecture enables scalability and real time processing of high velocity transaction streams. The predictive core was at baseline or better in FinTech tasks. The data quality pipeline was able to identify injected problems successfully.

Scenario walkthroughs showed that the LLM layer can explain technical concepts in plain language to non technical users. Every single processing is done in premise such that sensitive data does not leave the organization.

The **limitations** are that SDAF was only evaluated through technical benchmarks and scenario walkthroughs, not a real world field study. The walkthroughs were based on hypothetical users as opposed to real users. The study did not systematically compare different on premise LLM models or hardware configurations. The effects of adoption in the long term are not known.

Future research should deploy SDAF in real FinTech environments to study long term use and organizational impact. Controlled experiments with actual stakeholders including analysts, managers, and auditors are needed to evaluate how different groups perceive the LLM explanations. Future work should also compare performance and cost trade offs across various on premise LLM backends and explore alternative interaction designs.

Financial Big Data Management and Control and Artificial Intelligence Analysis Method Based on Data Mining Technology (Yang, 2022)

The **motivation** is that traditional manual financial work cannot keep up with the rapid growth of modern enterprises. Financial efficiency is in dire need of improvement in large organizations that should increase timeliness and quality of financial information. The technologies of artificial intelligence like UN mining of data, deep learning, and intelligent decision making provide an answer to mechanize routine operations in the financial sector and attain financial intelligence.

The **research problem** is that in the big data era, people have access to vast amounts of data but struggle to extract useful information for decision making. Conventional financial analysis techniques are also characterized by error, delay as well as failure to compute dynamic financial indicators. Current decision support systems do not have all functions to satisfy all the needs of users and thus some valuable business intelligence is being left unexploited.

The **aim** is to develop a data mining based financial analysis system that enables real time integration and analysis of business data. The system offers decision making recommendations and gives early warning of the risk to the management as well as acting as a guide to the other business ventures that are looking forward to modernize their finances.

The **scientific method** uses several techniques. Statistical analysis involves the use of Kolmogorov Smirnov tests, and t tests to filter the financial indicators, and the ones that have significant differences among populations are marked. CRISP DM data mining process organizes the analysis into three phases, problem identification and data warehousing, core analysis which converts data into knowledge model, and data output. The backpropagation trains neural networks by calculating prediction errors and adjusting weights to improve accuracy. The HIUA association rules algorithm has a hash structure to find patterns to identify financial risk and enhance speed and efficiency of access.

The **findings** show that eight financial indicators including asset liability ratio and return on assets follow normal distribution and show significant mean differences between populations. The size of assets is more suitable to forecast financial crisis in small businesses than the number of employees. The use of the prediction accuracy indicator and the inclusion of corporate governance indicators will be effective in minimizing the number of attributes required to analyze and predict and the rough set method to be applied. Early warning model, which is based on AI, is superior to logistic regression. The system was reported to have better bandwidth in the 7.625 Mbps compared with 3.125 Mbps on cloud based systems, in the experiment of fog computing, and the latency decreased as the number of sensing devices was increased.

The **limitations** include that the system functions are not comprehensive enough to serve all decision making needs. The algorithm does not have a flexibility to changing the support and confidence thresholds. Research was also confined to the Chinese companies listed that had suffered loss crises, which do not necessarily represent all types of enterprises or environment around the world.

Future research should improve data mining efficiency for large scale databases, expand methods for financial risk analysis and crisis early warning, and consider cloud data warehouses to reduce costs. For group enterprises, system implementation should follow step by step and cost benefit principles while considering the influence of human factors on successful operation.

Enhancing Financial RAG with Agentic AI and Multi HyDE: A Novel Approach to Knowledge Retrieval and Hallucination Reduction (Srinivasan et al., 2025)

The **motivation** is to improve accuracy and reliability in financial question answering, where errors can lead to significant monetary losses. LLMs have a high tendency to hallucinate and are able to process financial documents. The complexity, multi year and numerical density of financial text is a problem to Traditional Retrieval Augmented Generation systems.

The **research problem** is that current RAG systems fail at complex financial questions. Existing retrieval methods like HyDE confuse passages between annual reports of different years that differ only in critical numerical or temporal details. Non dynamic pipelines cannot handle multi step queries requiring calculations or verification. While knowledge graphs exist, they require expensive upfront processing costs, and current evaluation metrics do not reflect real world performance.

The **aim** is to present a new financial RAG model that improves retrieval and reduces hallucinations through two innovations. To begin with, Multi HyDE, a retrieval algorithm that does not diminish cover at the expense of higher token costs. Second, agentic AI pipeline to dynamically align multi step reasoning and utilize tools, such as web search and calculators, to complex queries.

The **scientific method** combines two approaches. Multi HyDE generates multiple hypothetical documents from diverse non equivalent queries rather than a single document, then uses dense vector search combined with BM25 keyword retrieval to improve coverage of structured information like tables. The agentic RAG pipeline uses an LLM as a coordinator. It first attempts retrieval, and if evidence is insufficient, it breaks the query into atomic steps, calls tools such as financial APIs and a Python calculator, and verifies intermediate findings before synthesizing a final answer.

The **findings** show that Multi HyDE with BM25 and rerankers significantly outperformed standard HyDE, CRAG, and LightRAG on key metrics including Recall, Factual Correctness, and Faithfulness. These advantages were attained without raising token costs over HyDE as well

as without the large initial expenses of graph based techniques. The agentic pipeline brought resiliency through providing more alternative routes of accessing relevant data.

The **limitations** include evaluation on a relatively small dataset, which may limit generalizability. The system cannot claim full deployment and is subject to hallucinations with more complicated dataset. Even high stakes applications still need human scrutiny and validation.

Future research should explore fine tuned Small Language Models using parameter efficient methods like LoRA for tasks such as query rewriting and hypothetical document generation, creating more lightweight solutions. Better evaluation metrics for financial RAG are needed, as current LLM based methods are unreliable for numerical answers and may incorrectly penalize precise responses or reward refusals to answer.

4.2 Comparative Analysis

The following section offers a comparative analysis of the presented state of the art, identifying and discussing the scientific gap that emerges from the five most recent research findings from 2022 to 2025. The analysis highlights key similarities and differences between the **objectives**, **methodologies**, **findings**, and **limitations** of the selected studies to better understand the current landscape and identify opportunities for further research.

The reviewed literature is also motivated by the growing need to apply artificial intelligence to the solution of real problems in finance. The **motivation** of Kong et al. (2024) is to emphasize that LLMs can process unstructured financial information in ways traditional models cannot. This is similar to Duane et al. (2025), who argue about the explosion of unstructured information from news, social media, and earnings calls those older methods struggle to process. The professional issue that motivates Al et al. (2025) is that many financial experts lack technical skills to apply highly sophisticated AI. The practical limitations of manual work in financial matters in large organizations inspire Yang (2022). Srinivasan et al. (2025) are driven by the risks of AI errors in high stakes financial decision making, particularly LLM hallucinations.

Although the studies share similar motivations their **objectives** differ. Kong et al. (2024) aim to construct a state of the art mapping of finance related LLMs. Duane et al. (2025) aim to review deep learning methods applied to unstructured financial data across text, audio, and images. Al et al. (2025) aim to construct a working system called SDAF designed for explainability and on premise data sovereignty. Yang (2022) aims to build a financial analysis and crisis early warning system using data mining. Srinivasan et al. (2025) aim to enhance financial question answering by minimizing hallucinations through Multi HyDE integrated with an agentic pipeline.

The **scientific methods** employed vary. Kong et al. (2024) and Duane et al. (2025) conducted literature reviews, with Kong classifying by model architecture and Duane by data type. The Design Science Research in Al et al. (2025) had an agile development. Yang (2022) used such statistical tests as K S and t tests, backpropagation and association rules algorithms. The study by Srinivasan et al. (2025) had an experimental design and developed Multi HyDE, which is a form of experimental data retrieval featuring key word based retrievals and an AI agent with the ability to break down multi faceted queries.

The **findings** across all studies demonstrate that AI significantly enhances financial analysis. Kong et al. (2024) discovered that domain specific models such as FinBERT achieve new standards, and smaller fine tuned models can be better than large models. Duane et al. (2025) demonstrated that deep learning models are more efficient in fraud detection and credit risk and multimodal models perform better with an AUC of 0.91 compared with numeric alone of 0.808. Al et al. (2025) have shown that SDAF satisfies all six design aspects such as explainability and data sovereignty. Yang (2022) has determined AI based early warning to be better than logistic regression. Srinivasan et al. (2025) demonstrated how Multi HyDE is superior to standard HyDE and CRAG on Recall, Factual Correctness, and Faithfulness with no rise in token costs.

The **limitations** reveal persistent challenges. According to Kong et al. (2024), there is the risk of lack of interpretability and hallucinations. According to Duane et al. (2025), there is a dearth of labeled data, black box models, non stationary data, and high computational costs. Al et al. (2025) admit that the evaluation was not performed based on real life deployment or benchmarking and walkthroughs. According to Yang (2022), there is the lack of scope to listed Chinese companies and the inflexibility of the algorithm. Srinivasan et al. (2025) acknowledge that the analysis is done on a limited set of data and the system needs to be controlled by human resources.

Future research directions converge on key themes. Kong et al. (2024) and Duane et al. (2025) point to multimodal learning and domain specific training. According to Al et al. (2025) and Srinivasan et al. (2025), this is deployed in real life scenarios and tested with real stakeholders. Yang (2022) encourages more research on the efficiency of data mining and data warehouses on the cloud.

Altogether, the reviewed articles confirm that AI offers great opportunities to enhance financial analysis, but only by overcoming transparency, data quality, and feasibility issues. Kong et al. (2024) and Duane et al. (2025) highlight multimodal learning and domain specific training as key directions. According to Al et al. (2025) and Srinivasan et al. (2025), systems must be easy to use and reliable to gain acceptance. Yang (2022) supports integrating statistical and machine learning solutions. The direction of future studies should focus not only on accuracy but also on explainable, adaptable, and regulation compliant AI systems effective in real world financial contexts.

<i>Article</i>	<i>Objectives</i>	<i>Methodology</i>	<i>Key Findings</i>	<i>Limitations</i>
Kong et al. (2024)	Evaluate LLMs for financial and investment management.	Literature review by model architecture.	Finance specific LLMs outperform general models. Small fine tuned models rival large ones. RAG boosts reliability.	Hallucination risk, lack of interpretability, and uncertain performance versus pre LLM benchmarks.
Duane et al. (2025)	Review deep learning methods for unstructured financial data.	Modality based literature review of deep learning architectures.	DL outperforms traditional methods in fraud detection and credit risk. Multimodal fusion exceeds numeric only. FinBERT excels in sentiment analysis.	Black box models limit regulatory acceptance. Labeled data is scarce. Models are sensitive to market regime changes.
Al et al. (2025)	Design an explainable, on premise LLM augmented analytics system (SDAF) for FinTech firms.	Design Science Research with agile prototyping and scenario walkthroughs.	SDAF meets all design requirements: high accuracy, fast processing, plain language explanations, and full on premise operation.	Scenario walkthroughs only; no real world deployment. No comparison of on premise LLM backends. Long term effects unknown.
Yang (2022)	Build a data mining driven financial analytics system for risk management and crisis early warning.	Statistical tests, CRISP DM, backpropagation neural network, and association rules algorithm.	AI based early warning outperforms logistic regression. Corporate governance data improves accuracy. Fog computing reduces latency.	Limited to listed Chinese firms. Algorithm lacks flexibility when confidence or support thresholds shift.
Srinivasan et al. (2025)	Improve accuracy and reduce hallucinations in AI based financial Q&A via Multi HyDE and an agentic pipeline.	Multi HyDE retrieval combining keyword search and an LLM agentic coordinator.	Multi HyDE outperforms standard retrieval methods on accuracy and faithfulness at no extra cost. Agentic pipeline handles complex queries.	Small evaluation dataset. Hallucinations persist on complex queries. Human supervision still required.

Table 1. Comparative Analysis

4.3 Scientific Gap

Based on the [Comparative Analysis](#) presented in the previous section, several persistent challenges were identified across the reviewed literature. While each of the selected studies contributes to advancing artificial intelligence in financial data analysis, a critical gap remains unaddressed. Existing research primarily focuses either on structured financial datasets or on qualitative analysis of unstructured data, but does not provide a unified solution capable of handling both simultaneously. In particular, no study proposes a framework that can accurately extract structured financial data embedded within unstructured corporate annual reports and present it in a structured and comparable output format.

Corporate annual reports inherently represent a hybrid data environment, combining structured numerical data such as financial statements and ratios with extensive unstructured textual narratives. This mixed format creates significant processing challenges, as structured financial data cannot be cleanly separated from the surrounding text without a system capable of understanding both forms at the same time (Duane et al., 2025). Although prior research demonstrates that integrating structured and unstructured data improves analytical performance, it does not provide a practical mechanism for identifying and extracting embedded numerical financial figures. Consequently, the core problem of reliably isolating and interpreting structured financial data from mixed format documents remains unresolved.

Another critical limitation identified in the literature is the reliance on large language models without adequately addressing their susceptibility to hallucinations. This issue is particularly significant in financial contexts, where even minor numerical inaccuracies may lead to incorrect conclusions and substantial financial risks (Kong et al., 2024). While retrieval augmented approaches aim to improve factual accuracy, they are primarily designed for qualitative question answering and do not address the extraction of precise numerical data (Srinivasan et al., 2025). Furthermore, these approaches remain experimental and continue to require human validation for complex financial queries. In addition, most existing solutions rely on cloud based architectures, which raises concerns related to data privacy and regulatory compliance, while existing On premise approaches do not address the extraction of structured financial data from mixed format documents and often assume that structured data is already available (Al et al., 2025; Yang, 2022).

As a result, the existing literature does not provide a comprehensive solution that addresses all aspects of this problem. There is no existing system that treats corporate annual reports as mixed format documents requiring structured data extraction, accurately extracts and presents numerical financial data in a structured and comparable format across one or multiple reports, enables interaction through a conversational interface, incorporates a hallucination mitigation strategy specifically designed for numerical extraction, and operates entirely within an On premise environment. Moreover, no On premise large language model or retrieval augmented generation system currently provides a reliable and deployable solution for this task.

Addressing this gap is of high practical importance for corporate environments. Financial data is highly sensitive and, in many cases, cannot be transmitted to external cloud services due to data privacy regulations and data sovereignty requirements (Al et al., 2025). At the same time, financial analysis within organizations continues to rely heavily on manual extraction of data from annual reports, which is time consuming, error prone, and difficult to scale (Yang, 2022). The increasing volume of unstructured financial information further intensifies this challenge and requires more advanced processing capabilities (Duane et al., 2025). A system capable of automatically extracting and structuring financial data within an On premise environment would significantly enhance efficiency, reduce human error, and enable reliable comparative analysis across multiple companies and reporting periods. In addition, minimizing hallucination in numerical outputs would improve the reliability of AI driven financial insights, which is essential in high stakes decision making contexts (Kong et al., 2024).

4.4 Research Question

The available research focuses predominantly on applying AI to pre structured financial datasets or on qualitative analysis of unstructured financial text, but no direct framework exists for extracting structured numerical financial data embedded within unstructured corporate annual reports and presenting it in a comparable, exportable format. Therefore, the research question derived from this gap is: "How can a Retrieval Augmented Generation system be designed to accurately extract structured financial data embedded within unstructured corporate annual reports and present it in a structured comparable output such as a spreadsheet or dashboard, while minimizing hallucinations in a fully on premise environment?"

5 Methodology Fundamentals

Since the main objective of this thesis is to develop a framework for analyzing corporate financial structured and unstructured data, This chapter is about the methods used to do the research for this thesis. It talks about something called Design Science Research Methodology and the steps that are part of it. The literature review is also used to support the methods used in this thesis. The research methods used here help create a foundation for designing the proposed model in a way and for evaluating the thing that is created. This thesis uses financial data to do this. The methods used in this thesis are important, for the financial data analysis.

Research in information systems can be done in ways. One way is Design Science Research. This approach is used by scholars like Offermann et al., Hevner and Peffers. It involves creating and testing IT solutions to fix problems in organizations and technology. This approach uses step-by-step guides like DM to help build and improve these solutions. Another way to do research is through methods. This involves talking to experts or observing them to understand how proposed solutions work in real-life situations. We also look at what has been researched. This is done through literature reviews, which help us understand what we already know. What we still need to find out. The next sections will explain how we used each of these research methods in this thesis. We will look at Design Science Research, research and literature reviews one, by one.

5.1 Basic Scientific Methodology

The Design Science Research Methodology is what I am using for this thesis. This is because it helps me to design and evaluate the IT thing I am making which's a model that uses RAG to get financial data from company reports that are not organized. I like the Design Science Research Methodology for this thesis because it gives me a way to do things from finding the problem to designing the thing and then checking if it works. The Design Science Research Methodology is good, for this because I am working with a model that needs to be used inside a companys system without using any outside services.

The literature review conducted in [Chapter 4](#) is embedded within the DSRM process, supporting the problem identification and solution design phases by establishing the state of the

art and informing the architectural decisions of the proposed model. All structural details of how DSRM is applied are discussed in [Section 5.2](#).

5.1.1 Design Science Research Methodology

This thesis will use Offermann et al. (2009), Hevner (2004), and Peffers et al. (2007) to define DSRM. These designs are used in the field of information system due to its structure to develop and evaluate innovative artifacts addressing practical problems within the field.

5.1.1.1 Offerman Design Science Approach

Offermann et al. (2009) propose a structured research process for Design Science in information systems, combining qualitative and quantitative research methods to ensure both relevance and rigour. As this entire section draws from the same source, all content presented hereafter refers to Offermann et al. (2009). The process is structured into three main phases: Problem Identification, Solution Design, and Evaluation, with a concluding step of Summarizing Results. The phases are not strictly sequential and often refer back to each other throughout the research process.

1) Problem Identification

defining something that's a real issue in the information systems field. We do this in four steps. First we find the problem. Then we do some research to see what people already know about it. What is still not solved. Next we talk to experts who work in the field to make sure the problem is really important. Finally we think about why this problem's relevant and come up with a general idea of how we might solve it. We keep working on this idea throughout our research.

2) Solution Design

is about creating something that solves the problem we found. We do this in two steps. First we design the solution using our knowledge and what other people have already figured out. Then we do some research to learn about methods and solutions that can help us make our solution better. We write down all our decisions so we can see how we got to our solution.

3) Evaluation

is about testing it to see if it works. We start by making our general idea more specific. Then we test it in ways like doing case studies or working with people, in the

field. We also ask experts what they think. We do experiments to compare our solution to what already exists.

- ✦ Results are summarized and published, with the possibility of publishing intermediate results to gain earlier feedback and improve the overall research outcomes.

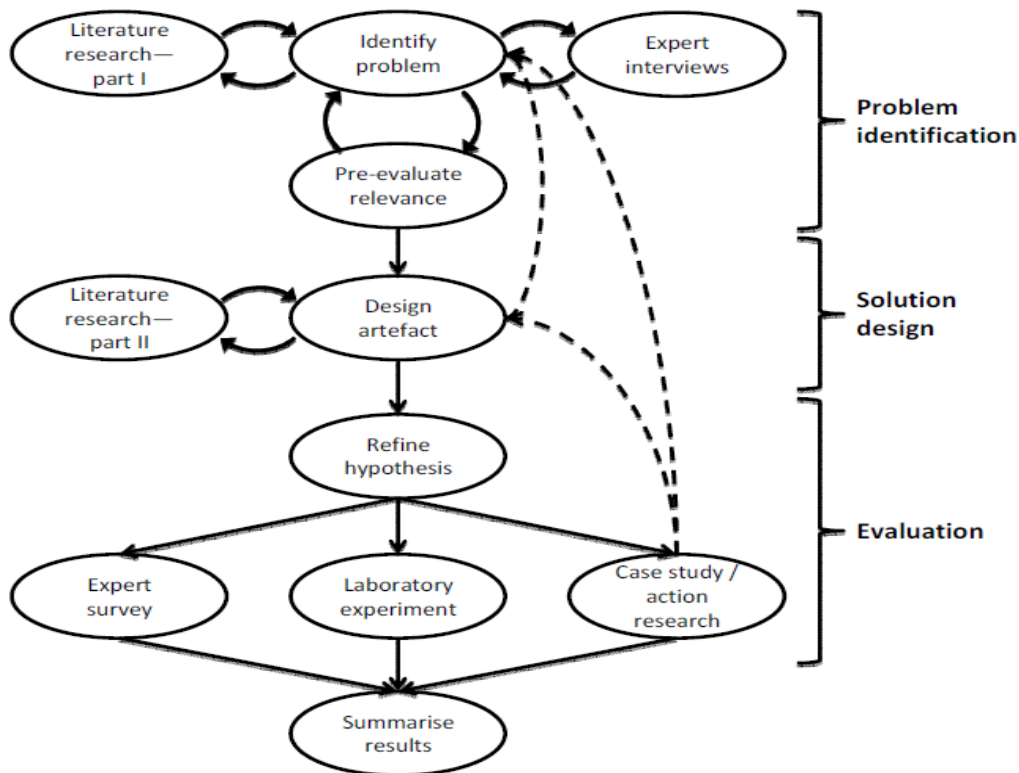


Figure 1. Proposed Research Process by Offermann et al., 2009.

5.1.1.2 Hevner Design Science Approach

Hevner et al. (2004) established a foundational framework for Design Science Research (DSR) in Information Systems, structured around three interconnected cycles and seven guiding principles.

1) Relevance Cycle

The Relevance Cycle helps us understand the business environment and its needs. It looks at the company's goals and the technology it already uses. We also get feedback from testing our ideas, which helps us make sure our research is solving problems.

2) Rigor Cycle

This cycle connects our research to what experts know. We use tested methods and theories to build and test our ideas. This way we make sure our solutions are based on knowledge and add new ideas to help the field grow.

3) Design Cycle

This cycle in the Design Cycle we build and test our ideas step by step. We create an IT solution, like a new method or tool and test it thoroughly. We use ways to test it like simulations or experiments until it solves the problem.

- ✦ The framework gives us seven rules to follow. We must create a solution that solves a problem for a company focusing on technology solutions, for business needs. We must show that our solution works well and is useful. We must use methods to build and test our solution. We should design it in a way that uses what we have to get what we want considering the company's limits. Finally we must explain our findings in a way that both tech and business people can understand.
- ✦ Together, these cycles and guidelines balance relevance, rigor, and design. These cycles and rules help us balance practicality, scientific knowledge and innovation. The Relevance Cycle keeps us focused on needs the Rigor Cycle grounds us in expert knowledge and the Design Cycle helps us create new solutions through testing and iteration. This way we get results that're both useful and scientifically sound.

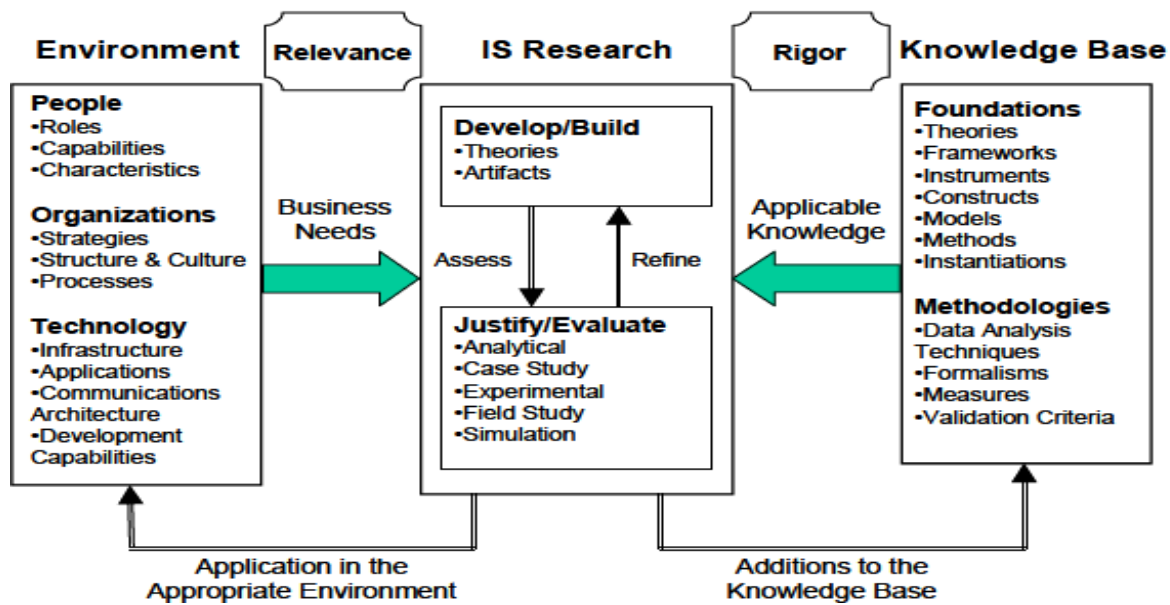


Figure 2. Proposed Research Process by Hevner, 2004

5.1.1.3 Peffers Design Science Approach

Peffers et al. (2007) came up with a way to do design science research in Information Systems. This is a deal because it fills a gap that has been around for a long time. They called it the Design Science Research Methodology. It helps people do research and present their findings in a way. The Design Science Research Methodology has six steps that people can follow. These steps are:

Problem Identification and Motivation: the person doing the research figures out what the problem is and why it is important to solve it. If the problem is not well defined it is hard to make something that will actually work.

- 1) **Defining the Objectives of a Solution** the researcher thinks about what a good solution would look like based on the problem and what they already know.
- 2) **Design and Development** this is where the researcher actually makes the thing they are working on like an idea, a model or a method.
- 3) **Demonstration:** the researcher tries out the thing they made to see if it works and if it is useful.

- 4) **Evaluation:** the researcher checks to see how well the thing they made works. If it needs to be improved.
 - 5) **Communication:** the researcher shares what they found out with people, like academics and professionals so everyone can learn from it (Peffers et al., 2007).
- ✦ One of the things, about the Design Science Research Methodology is that it is flexible. This means that researchers can start with the problem or the solution or even just an idea. They can start wherever they want depending on what they're trying to do and it works in lots of different situations.

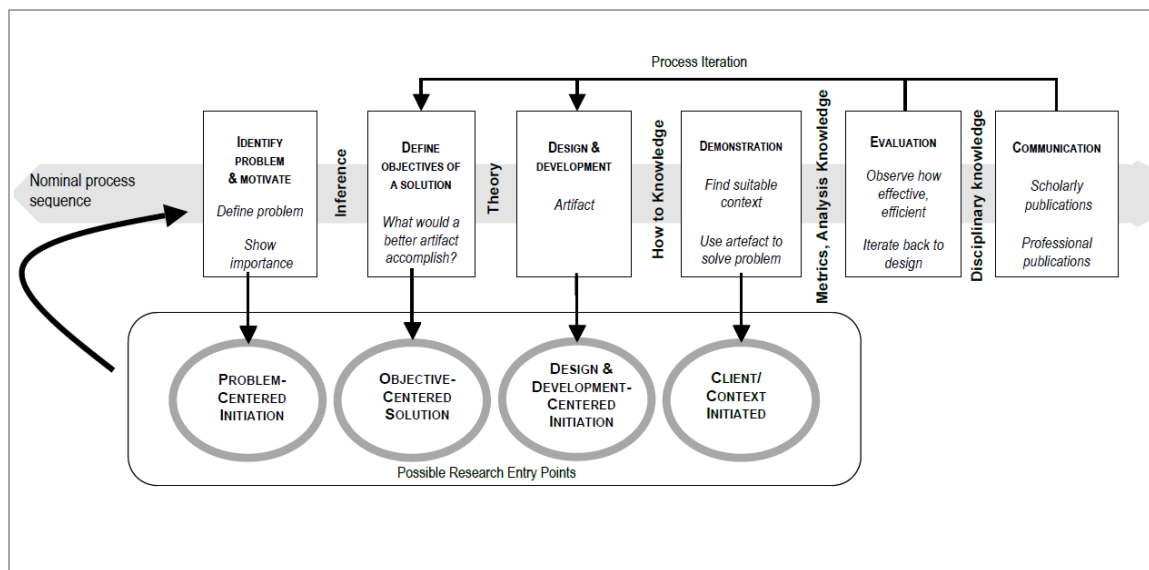


Figure 3. Proposed Research Process by Peffers, 2007.

5.1.2 Qualitative Interviews

Qualitative research uses non numerical data like interviews, observations and documents to understand issues in their natural setting. This approach helps to explain phenomena by looking at peoples experiences and surroundings.

Unlike methods that focus on numbers and statistics qualitative research is about understanding and interpreting peoples experiences. It tries to find out what people think and feel about their experiences. The method is flexible. Often uses a simple logic to make sense of the data. Researchers collect data through stories, observations, conversations and texts. They then analyze the data by grouping ideas together and looking for common themes.

In the field of Information and Communication Technology qualitative research has become more popular. This is because the field now looks at how technology affects people and organizations not technical issues. Qualitative research is good for studying how technology interacts with people and institutions.

Qualitative research helps us understand peoples experiences with technology. It looks at how people use technology in their lives. Qualitative research is useful, for understanding issues related to technology(Crozier et al., 1994).

5.1.2.1 Delphi Method

The Delphi Method is a way to gather expert opinions through a structured and repeated process. It's part of qualitative research, which means it focuses on understanding complex situations that can't easily be measured with numbers. Basically, it relies on what experts think and interpret about various issues (Massaroli et al., 2017).

This method involves several rounds of consultations with a group of carefully chosen experts. They're selected not because they represent a larger group but because they have deep knowledge in their fields. One of the key aspects of the Delphi Method is that after each round, the responses from participants are shared with everyone involved. This allows each person to think about the group's overall views and either change their opinions or stand by them. Throughout this process, the anonymity of the participants is maintained so no one voice dominates the discussion.

The process starts with open-ended questions, usually in the form of discussions, letting people share their thoughts freely. Then, the responses are analyzed to find common themes and patterns. After that, more structured methods come into play, using rating scales and statistics to see how much agreement there is among participants. This approach shows that it can blend different research methods, but it really leans towards the qualitative side. It's especially useful in research areas where there isn't much existing theory, and where having informed opinions from experts can lead to solid and meaningful results.

5.1.2.2 Technology Organization Environment Framework

The The Technology, Organization, and Environment (TOE) framework, which was first put together by Tornatzky et al., 1990, gives us a way to look at what influences how organizations decide to adopt new technology. It breaks things down into three parts. First, there is the technological aspect. This looks at what an organization already has in terms of IT like their systems, hardware, software, and how ready they are to bring in and support new tech. Then, we have the organizational side, which focuses on the internal factors that can affect adoption. This includes things like how big the firm is, its structure, the resources available, and whether the culture is open to change. Lastly, there's the environmental dimension. This takes a look at what's going on outside the organization, considering all the external pressures and conditions that can impact decisions like industry standards, regulations, and competition. All these factors can really influence how quickly and effectively a company decides to adopt new technologies.

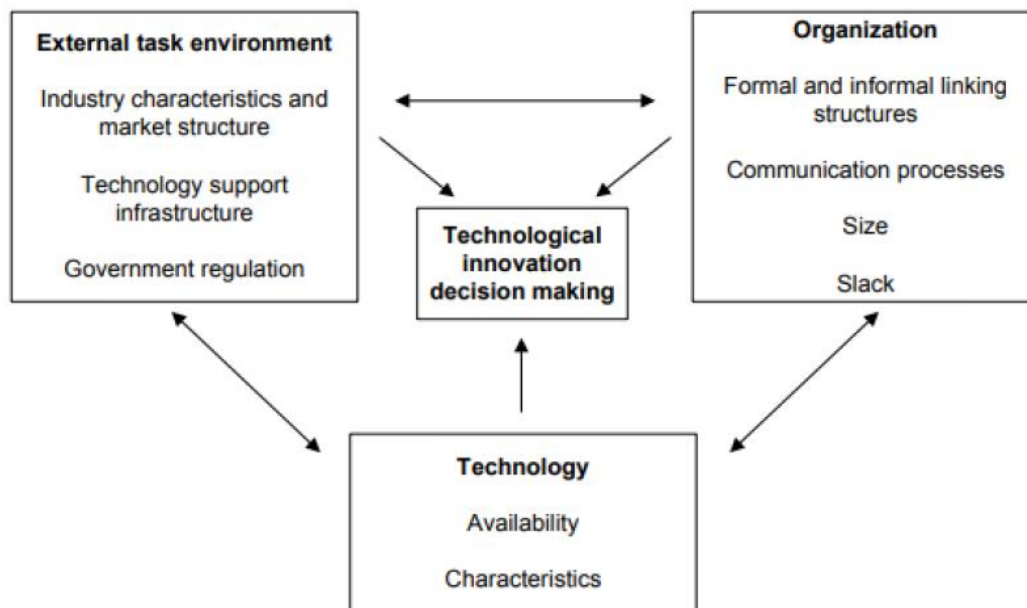


Figure 4. TOE Framework adopted from Tornatzky et al., 1990.

This framework highlights the factors that could impact one another and helps in making interview questions that allows this research to validate the viability of the framework to be built.

5.1.3 Literature Review

A literature review is an approach in conducting research where there is a detailed analysis and integration of the state of the art about a particular concept, theory, and methodology in order to build on the basis of research field and discover the areas where more research is required (Paul & Rialp Criado, 2020). Not only does it help in analyzing the body of knowledge available in a particular area of research, but it also helps in providing guidance towards future research directions by exploring the uncharted waters of theories, concepts, methods, and contexts. In this particular research, a literature review was carried out as an integral part of the problem identification process, where a review of literature was undertaken to explore the existing body of knowledge related to the research topic, find the underexplored research gap, and define its significance.

A systematic review of literature may generally be classified into one of four types domain based, theory based, method based or meta analytic according to its unique functions in relation to the type of research question posed. Whatever the type chosen, a good literature review is supposed to follow certain guidelines which include selecting an original and underexplored topic, clearly stating the criteria and period of journals chosen for the review, using appropriate keywords along with criteria of inclusion and exclusion for identifying relevant articles, and stating the research gaps clearly and presenting a future research agenda. In the present research, such principles were followed in the problem statement phase.

5.1.4 Data Science Research Methodology

Data science is a body of principles and techniques for applying analytical methods to data at scale, with the primary objective of building predictive models from probabilistic correlations (Brodie, 2018). This predictive orientation is operationalized through the Cross Industry Standard Process for Data Mining (CRISP DM), which is discussed in further detail in [Section 5.1.4.1](#).

5.1.4.1 Cross Industry Standard Process for Data Mining Framework

The CRISP DM framework can be understood as a structured and iterative approach to designing data mining models, positioning model development itself as a form of design activity. It consists of six interconnected phases that guide the transformation of raw data into actionable

insights. The process begins with business understanding, where objectives are defined from an organizational perspective and translated into a data mining problem. This is followed by data understanding and data preparation, which focus on collecting, exploring, and refining data into a usable format. The modeling phase then applies analytical techniques to extract patterns, while evaluation ensures that the developed model aligns with the initial business objectives. Finally, deployment operationalizes the model within a real world context.

A key characteristic of CRISP DM is its iterative nature, allowing movement back and forth between phases to refine both the problem and the solution. Unlike traditional design approaches that emphasize problem definition, CRISP DM is more solution oriented, placing stronger focus on developing and assessing models based on existing data and techniques. This reflects its practical orientation, where continuous feedback and adaptation are essential for improving model performance and relevance. As a result, CRISP DM not only structures the technical workflow of data mining but also supports decision making by aligning analytical outputs with business needs (Kannengiesser & Gero, 2023).

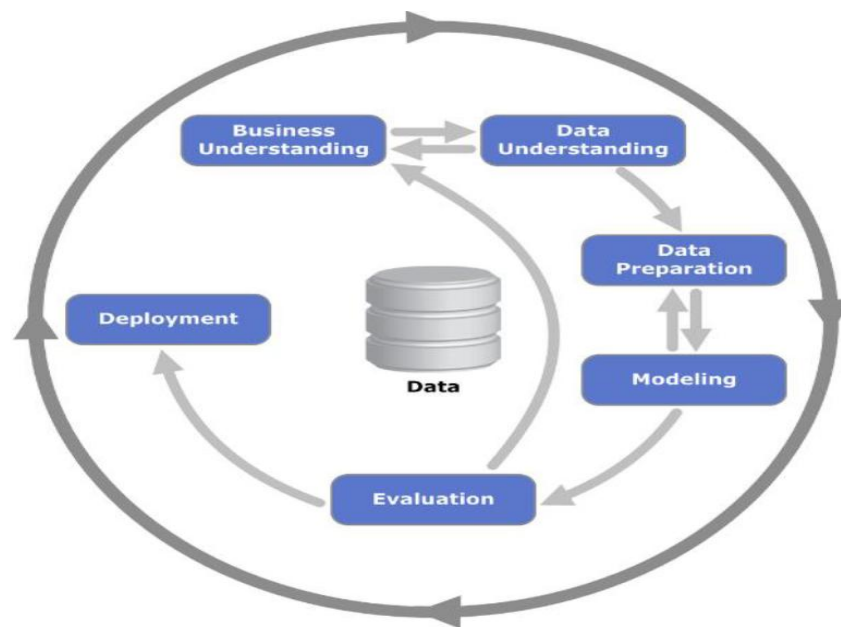


Figure 5. Proposed Research Process by Kannengiesser & Gero, 2023

6 Solution Approach

This section presents the solution approach for all the sections previously executed in theory as it is adapted from Offermann et al. (2009), Hevner (2004), and Peffers et al. (2007), tailored to the design of a RAG based framework for extracting structured financial data from unstructured corporate financial reports in a fully on premise environment. Since the core contribution is the development of a framework, DSRM serves as the primary methodology, complemented by the literature review and data science through CRISP DM in supporting roles. Each phase produces a clear output serving as the direct input to the next, ensuring a traceable flow from problem identification through to the communication of results.

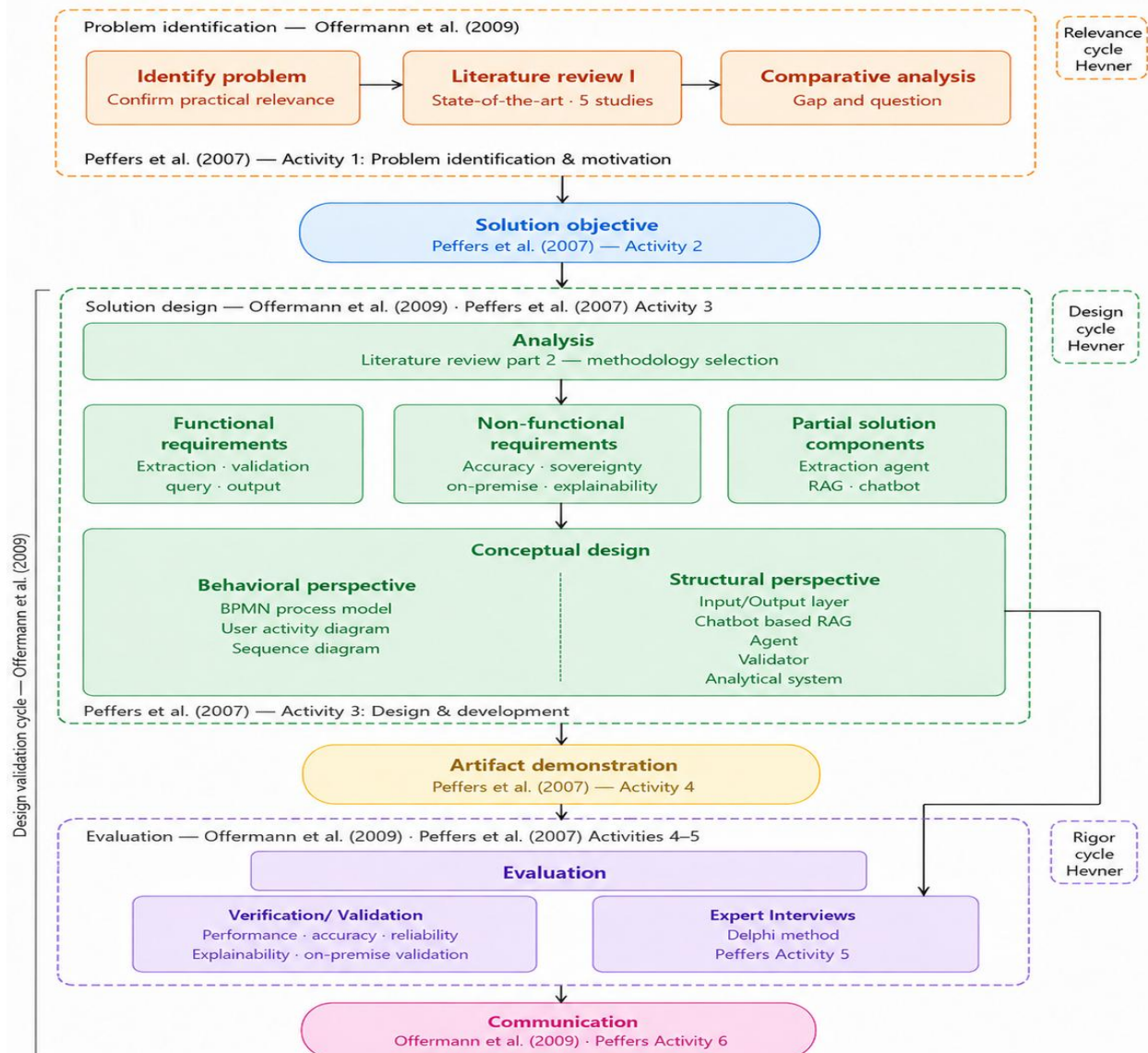


Figure 6. Solution approach adapted from Offermann et al. (2009), Hevner (2004)

The process begins in the [problem identification](#) phase following [Offermann et al. \(2009\)](#), where the research problem is identified, a first [literature review](#) is conducted across five studies, and a comparative analysis yields the scientific gap and research question. This corresponds to Activity 1 of [Peffer et al. \(2007\)](#), while [Hevner's \(2004\)](#) relevance cycle ensures the problem remains grounded in the real world context of financial data processing and data sovereignty. The solution objectives are then derived in correspondence with Activity 2 of Peffer et al. (2007), feeding into the solution design phase.

[The solution design](#) phase corresponds to Offermann et al.'s (2009) second phase and Activity 3 of Peffer et al. (2007). An analysis stage opens the phase as a second literature review on methodology selection, where CRISP DM is adopted as the guiding data science process model complementing DSRM with a practical data oriented workflow. [Functional requirements](#) and [non functional requirements](#), addressing extraction accuracy, data sovereignty, on premise deployment together with [partial solution](#) components including the extraction agent, chatbot based RAG layer in the [conceptual design](#). This is expressed through a [behavioral perspective](#) via BPMN models, activity diagrams, and sequence diagrams, and a [structural perspective](#) defining the input/output layer, chatbot based RAG, agent, and validator layers to execute the module. Hevner's (2004) design cycle governs this phase through continuous iteration between the problem, design decisions, and knowledge base.

In the evaluation phase, the artifact is demonstrated through Activity 4 of Peffer et al. (2007) and evaluated through Activity 5 via expert interviews designed around the Delphi method, as detailed in [Section 5.1.2](#), enabling iterative consultation with domain experts until agreement on the framework's utility is reached. The design validation cycle of Offermann et al. (2009) allows iteration back to the solution design phase if needed, while Hevner's (2004) rigor cycle ensures evaluation against established information systems criteria. The phase concludes with the summarisation and communication of results corresponding to Activity 6 of Peffer et al. (2007).

6.1 Problem Identification

This section fulfills the problem identification phase of Offermann et al. (2009) and Activity 1 of Peffers et al. (2007) through three components: [Section 6.1.1](#) that identifies the core problem, [Section 6.1.2](#) the supporting literature review, and [Section 6.1.3](#) the comparative analysis, both conducted in [Chapter 4](#), that together establish the scientific gap and research question driving this research.

6.1.1 Identifying the Problem

[The literature review](#) and [Comparative analysis](#) identified in previous chapters a key gap in AI driven financial analysis, particularly in handling corporate annual reports that combine structured financial data with unstructured textual content. Existing approaches largely rely on pre structured datasets or focus on qualitative tasks, failing to support accurate extraction and utilization of numerical financial data. Additionally, current decision support systems lack the ability to integrate structured and unstructured data into a unified framework, limiting their effectiveness in real world corporate use. The use of LLMs, even with RAG, remains constrained by hallucination risks, which are especially critical in financial contexts where accuracy is essential.

Moreover, many existing solutions depend on cloud based architectures, raising concerns regarding data privacy and regulatory compliance. On premise alternatives remain limited in capability and practical implementation. Therefore, the core problem is the absence of a reliable system that can extract, structure, and analyze financial data from annual reports while ensuring accuracy, comparability, and secure on premise deployment this problem identification is explain thoroughly in [section 1.2](#).

6.1.2 Literature Review

As established in [Section 4.1](#), the state of the art across five selected studies from 2022 to 2025 demonstrates that no existing work provides a unified solution capable of extracting structured numerical financial data from mixed format corporate annual reports, presenting it in a comparable output format, enabling conversational querying, mitigating hallucinations in numerical extraction, and operating entirely on premise.

6.1.3 Comparative Analysis

As established in [Section 4.2](#), the comparative analysis of the five reviewed studies across objectives, methodologies, findings, and limitations confirms that the gap identified in [Section 6.1.1](#) is consistent across independent works, and directly informs the solution objectives defined in [Section 6.2](#) and the design decisions presented in [Section 6.3](#).

6.2 Solution Objective

Following [peffers](#) the primary objective of this research is to design and develop a model that leverages Retrieval Augmented Generation and large language models to extract structured financial data embedded within unstructured corporate annual reports, transform it into a structured and comparable format, and enable fact based conversational querying, all within a fully On premise environment.

Corporate financial data analysis continues to rely heavily on manual extraction and processing of financial figures from annual reports, which combine structured elements such as income statements and balance sheets with unstructured textual narratives. This mixed format makes automated and accurate extraction a complex and unresolved challenge (Duane et al., 2025; Yang, 2022). While large language models have demonstrated strong capabilities in natural language understanding and conversational interaction, they are not inherently reliable for precise numerical data extraction due to the risk of hallucinations, which presents significant limitations in high stakes financial contexts (Kong et al., 2024; Srinivasan et al., 2025).

In response to these challenges, this research aims to develop a solution that not only utilizes the strengths of large language models and Retrieval Augmented Generation, but also addresses their limitations through a structured and controlled approach to data extraction and validation.

To achieve this, the objectives of the proposed solution are as follows:

1. To review and analyze existing methodologies and architectures that utilize large language models, Retrieval Augmented Generation, and artificial intelligence based data extraction techniques within corporate financial data analysis, with a particular focus on structured data extraction from mixed format documents such as annual reports.
2. To design a model capable of processing corporate annual reports and parsing their content in order to identify and isolate structured financial data, including income statements, balance sheets, and financial ratios, from within unstructured document formats.
3. To develop a mechanism that ensures the accuracy and reliability of the extracted financial data by minimizing hallucination risks associated with large language models, particularly in the context of numerical data extraction.
4. To transform the extracted financial data into a structured and standardized format that enables comparability across multiple reports, companies, and reporting periods, supporting more effective financial analysis.
5. To integrate a conversational interface that allows users to query the extracted financial data in a fact based manner, ensuring that responses are grounded in verified data rather than generative assumptions.
6. To ensure that the entire solution operates within an On premise environment in order to maintain data privacy, regulatory compliance, and data sovereignty, which are critical requirements in corporate financial contexts (Al et al., 2025).

6.3 Solution Design

This section translates the problem identified in [Section 6.1](#) and the objectives defined in [Section 6.2](#) into a concrete framework design, organized into two parts. [Section 6.3.1](#) presents the Analysis, covering the Functional Requirements, Non Functional Requirements, and Partial Solutions. [Section 6.3.2](#) presents the Conceptual Design across the Behavioral Perspective and the Structural Perspective.

6.3.1 Analysis

Following [Peffer](#) et al. (2007), the solution design phase opens with an analysis stage that serves as a second literature review, distinct in purpose from the state of the art conducted in [Chapter 4](#). While the first literature review was oriented toward problem identification, mapping the existing landscape of AI driven financial data analysis and extracting the scientific gap that grounds this research, this second review is solution oriented. Its objective is to investigate the specific architectural components, technical approaches, and implementation strategies that are most relevant to the design of the proposed framework.

Alnefaie et al. (2021) introduce a chatbot, which is considered to be the technical approach that utilizes natural language processing and natural language understanding in order to understand user queries. The dialogue manager identifies keywords from the user input and sends these keywords to the knowledge engine, which identifies the type of the question and finds its answer in the knowledge database. The classification algorithm described above, which identifies different question types depending on their difficulty level, serves as the basis for the question classification phase proposed by the authors, where the bot should identify whether the user's question can be directly answered from the document or needs analysis.

According to Du et al., 2025, there is a RAG technique for scientific papers in PDF form which does not require a relational database. This technique uses a processing pipeline that splits the documents into three components: text fragments, tables transformed to structured CSV data using few-shot prompting, and figures processed with a vision-language module. All components are represented as text summaries with a large language model, embedded into a vector, and saved in a vector database. To perform retrieval, the query is encoded, followed by the similarity search, and then the retrieved text and table schema are processed by the large language model to create a structured data table based only on the paper. As such, it is possible to create a complete extraction pipeline solely based on unstructured PDF documents without any enterprise system support.

According to Alenezi (2025), agentic AI refers to an AI system that has the ability to recall previous encounters, devise a strategy to accomplish its objective, utilize instruments such as computation, learn from errors, and keep going until the job is finished. As opposed to a straightforward chatbot that answers one question at once, an agentic system can independently perform multi-step tasks. The principle of agentic AI design serves as the guiding philosophy behind the agent in our recommended architecture, which processes the extracted information from the chatbot-based RAG system when an analytic question is identified.

Odobesku et al. (2025) introduce the concept of nanoMINER, which is a multi-agent technical approach aimed at automatically extracting structured data from scientific literature. The ReAct agent coordinates with specific agents such as a named entity recognition agent that helps extract entities from the text and a vision agent that helps understand figures and non-standard tables. All extracted information is combined into structured JSON or tabular format based only on the information found in the documents. This multi-agent extraction methodology is applied by the agent when structuring the extraction of data based on their proposed system by which financial data present in the uploaded PDF file are extracted.

According to Ma et al. (2025), a validator is a piece of software used to check if the provided input fulfills certain requirements. Thus, the validator provides a binary output on whether the requirements were fulfilled before proceeding with any further actions. In our case, the validator serves as a gatekeeper which verifies the validity of the information provided before proceeding with further actions with the input information. The concept described above directly leads to the development of the validator stage of our architecture since it involves checking the figures that have been extracted from the provided CSV against their initial locations in the source PDF uploaded by the user.

As per Miles and Mauskopf (2024), there is a multi-step process of generating a structured output where the first step is defining the information requirements, followed by creating a standardized template and generating a searchable and structured output along with the necessary instructions for the users. The structured output framework serves as the basis for the last phase of the proposed framework where the validated CSV data is stored in an Excel spreadsheet, which acts as both the data storage and output medium. Based on the request from the user, a dashboard is created automatically, either when the user defines the required visualization method or simply asks for the analysis.

6.3.1.1 Functional Requirements

Building on the insights gained from the solution analysis and the partial solutions identified in the preceding section, the proposed framework shall fulfil the following functional requirements:

1. Multi Modal Document Parsing: The system will divide the financial PDF documents from corporations into three sections: text, financial data tables that will be transformed into CSV, and graphs which will be analyzed using vision language, working solely on the original document without relying on any other databases or warehouses. (Du et al., 2025).

2. Natural Language Query Classification: The system shall receive user questions through a conversational chatbot interface and classify each query as standard or analytical. notifying the user within the chat if the requested information is not present in the uploaded document. (Alnefaie et al., 2021)

3. Direct Document Response: The system shall answer standard queries directly from the PDF content and return a response to the user within the conversational interface without any extraction pipeline.

4. Agentic Financial Data Extraction: Upon detection of an analytical query, the system shall activate an agent that autonomously extracts the relevant financial figures from the parsed document and structures them into a CSV file for validation. (Alenezi, 2025; Odobesku et al., 2025)

5. Binary Validation Against Source Document: The system shall validate every figure in the extracted CSV against its original location in the uploaded PDF, forwarding verified data to output generation upon success, or notifying the user and terminating the process upon failure. (Ma et al., 2025)

6. Automated Dashboard and Structured Output Generation: The system shall load validated data into an Excel file and automatically generate a dashboard tailored to the user's query, whether a specific visualization format is requested or not, returning both the Excel report and the raw CSV to the user as downloadable outputs through the chat interface. (Miles and Mauskopf, 2024)

7. Integrated Structured and Unstructured Data Processing: The system shall simultaneously handle both the structured numerical figures and the unstructured textual narratives present within a single corporate financial document, treating the annual report as a unified hybrid source rather than processing each data type in isolation.

8. Hallucination Constrained Numerical Extraction: The system shall structurally prevent large language model hallucination in numerical outputs by ensuring the LLM operates only over data that has passed binary validation, never reasoning directly over raw document content when generating structured financial figures.

9. Conversational Output Delivery: The system shall deliver all outputs, including direct responses, validation notifications, and downloadable files, exclusively through the conversational chat interface, maintaining a single unified interaction point for the user throughout the entire process.

10. On Premise Deployment: The system shall operate entirely within an on premise environment with no data transmitted to any external cloud service or third party API at any stage, ensuring compliance with data sovereignty requirements and financial data privacy regulations applicable to corporate environments. (Al et al., 2025)

6.3.1.2 Non Functional Requirements

Building on the functional requirements defined in the preceding [section 6.3.1.1](#) the proposed framework for RAG based structured financial data extraction from corporate annual reports shall additionally satisfy the following non functional requirements:

1. Extraction Accuracy: The system shall maintain a high level of numerical precision across all extracted financial figures, ensuring that values identified from the uploaded financial document are correctly isolated and faithfully represented in the structured CSV output, with any inaccuracy detectable and rejected by the validation component before output generation proceeds.

2. Hallucination Containment: The solution must be designed in such a way that it will inherently avoid large language model hallucination on numbers through the process of making sure that the LLM is solely trained using data that passes binary validation from the source document.

3. Document Only Operation: It is to be understood that the proposed system will run purely on the user's provided PDF document as the basis for its source of information without having to require any kind of relational databases, data warehouse, ERP systems, or any other type of enterprise data systems, irrespective of the phase of the pipeline process. (Du et al., 2025)

4. On Premise Data Sovereignty: The system shall ensure that no financial data, document content, or extracted figure is transmitted to any external server, cloud API, or third party service at any point during operation, with all processing occurring entirely within the organization's own infrastructure. (Al et al., 2025)

5. Transparency: The system shall present outputs in a manner that allows users to trace each extracted financial figure back to its origin within the source document, ensuring that financial analysts can verify the basis of any generated insight and maintain confidence in the outputs produced.

6. Reliability: The system must exhibit consistent and reliable behavior throughout the whole pipeline flow beginning from document intake up to classification, extraction, validation, and

production of outputs in such a way that should any failure happen during processing, it will be clearly conveyed to the user via the conversation interface.

7. User Friendliness: The conversational interface design should ensure that users can upload documents, ask questions, receive responses, and download output from the system all within a single conversation window without the need for technical knowledge of the extraction process.

8. Modularity and Extensibility: The framework is designed to be structured using a modular structure where the various modules like the parser, classification module, extractor, validation module, and output generator module can each be upgraded, maintained, or substituted individually without impacting any other part of the pipeline process.

6.3.1.3 Partial solution

Based on the solution analysis conducted in the [section 6.3.1](#), the following partial solutions have been identified from existing literature to serve as the architectural building blocks of the proposed framework. Each partial solution addresses a specific component of the design and is directly traceable to a source reviewed in the analysis phase.

The first building block refers to the handling of corporate financial PDFs containing narrative text, financial tables, and figures in one mixed-source document. As shown by (Du et al., 2025), a RAG approach is able to parse the content of the document in three different modes – text, tables, represented as CSV data structures, and figures – working with the unstructured PDF without connecting it with any database or enterprise backend services. Hence, the technical basis is provided for the document parsing phase.

The second building block focuses on the intelligent handling of queries from users via the chatbot application. According to Alnefaie et al. (2021), conversational agents should be able to classify user questions according to their types and route every question to the relevant response pathway. The described concept forms the basis for the use of a chatbot based RAG model, which is capable of classifying user questions into standard, analytical, or irrelevant categories and processing each of them appropriately.

The third building block addresses the extraction and structuring of financial data for analytical queries. Alenezi (2025) describes agentic AI as a system capable of autonomously forming plans, using tools, and completing multi step tasks without human intervention. Odobesku et al. (2025) complement this with a multi agent architecture that extracts structured data from document content and consolidates outputs into tabular format. Together, these inform the agent component of the proposed framework, which activates upon detection of an analytical query, extracts the relevant financial figures, and organizes them into a structured CSV file.

The fourth building block addresses the hallucination risk associated with large language model numerical outputs. Ma et al. (2025) describe a validator that applies a binary verification mechanism to confirm whether extracted data satisfies a defined set of constraints before further processing proceeds. This directly informs the validator component of the proposed framework, which cross checks every figure in the extracted CSV against its original location in the source PDF, forwarding verified data to output generation upon success or notifying the user and terminating the process upon failure.

The fifth building block addresses the generation of structured and interpretable outputs. Miles and Mauskopf (2024) present a structured output process that produces a standardized, user guided analytical tool from defined information needs. This informs the final stage of the proposed framework, in which validated data is loaded into an Excel file where a dashboard is automatically generated in response to the user's query, and both the Excel report and the raw CSV are returned to the user as downloadable outputs through the conversational interface.

6.3.2 Conceptual Design

The conceptual design section translates the defined requirements into a concrete representation of the proposed framework across two complementary perspectives. The behavioral perspective, presented in [Section 6.4.1](#), describes the framework's runtime process flow and decision logic. The structural perspective, presented in [Section 6.4.2](#), defines its core components and layered architecture.

6.3.2.1 Behavioral Perspective

The proposed framework operates through a Chatbot Based Retrieval Augmented Generation (RAG) system designed to process corporate financial documents uploaded by the user. The process begins when the user uploads a financial document in PDF format and submits a question through the chatbot interface. The Chatbot Based RAG then classifies the user's question into one of two categories: a standard query or an analytical query.

A standard query refers to a question that can be directly answered from the uploaded document without requiring additional analytical processing, such as retrieving total revenue or operating income figures. For standard queries, the Chatbot Based RAG extracts the relevant information directly from the PDF and returns a direct response to the user through the chatbot interface. In case of an out of scope query, it refers to a request for information that does not exist within the uploaded PDF document and the process terminates.

In contrast, an analytical query refers to a request requiring additional financial analysis, calculations, comparisons, aggregations, or SQL like operations that cannot be directly answered through simple retrieval. When an analytical query is identified, the Chatbot Based RAG extracts the relevant financial figures from the uploaded PDF and forwards the extracted data to the Agent layer. The Agent structures the extracted figures into a CSV file and passes the file to the Validator.

The Validator functions as the hallucination mitigation mechanism of the framework by cross checking the extracted CSV data against the original uploaded PDF to ensure that the extracted values are accurate and consistent with the source document. If the validation fails, the system notifies the user that re uploading the document is required, and the process terminates. If the validation is successful, the validated CSV file is loaded into the Analytical System, where the user's analytical query is executed. The Analytical System then generates a dashboard or visualization presenting the analytical results in a structured format, which is subsequently displayed to the user through the chatbot interface, concluding the process illustrated in the BPMN (Business Process Model Notation) model in figure 7.

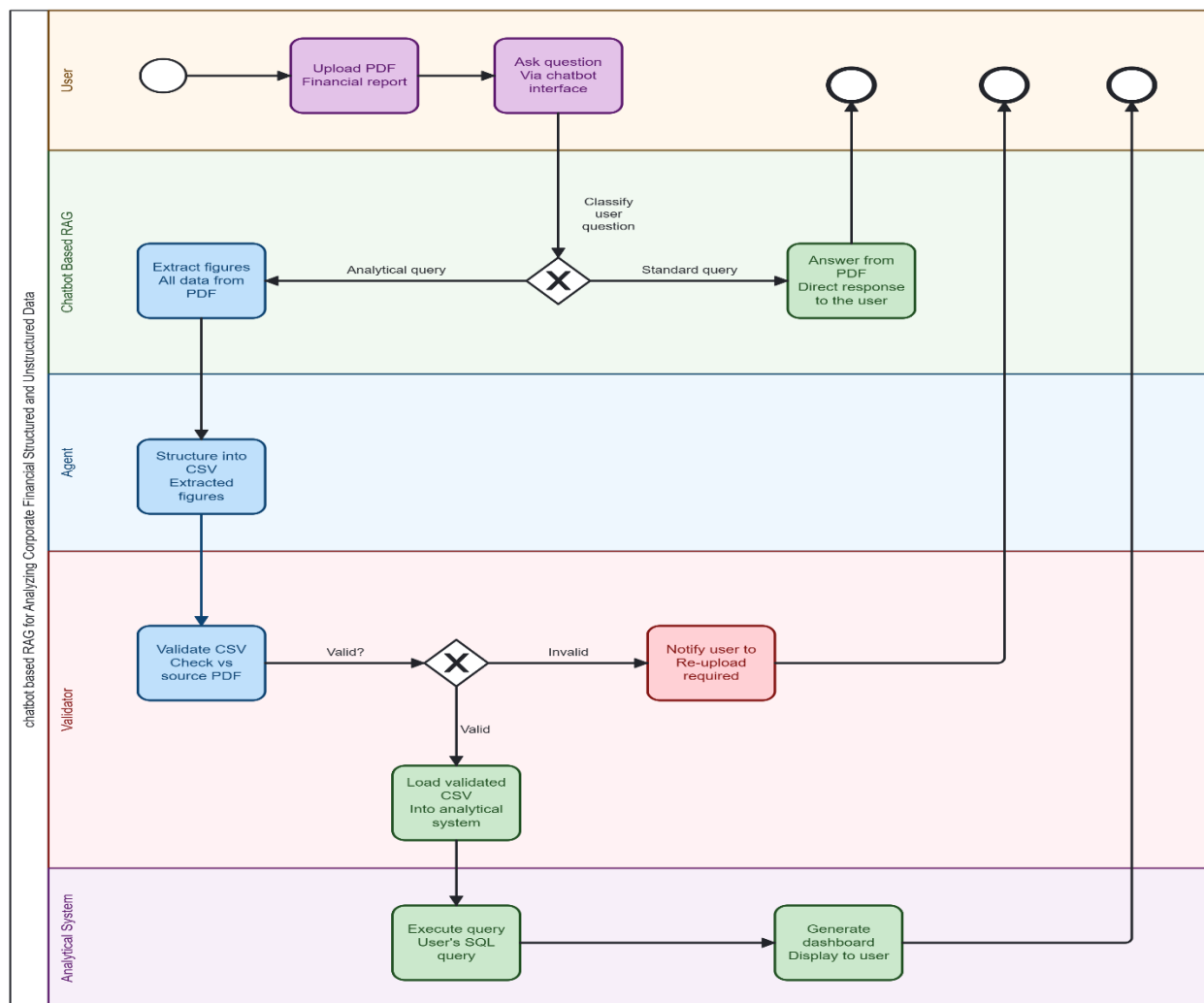


Figure 7. BPMN Workflow of Chatbot Based RAG for Analyzing Corporate Financial Data

The sequence diagram in Figure 8 organizes the framework into four layers: User Layer, Chatbot Based RAG Layer, Agent Layer, Validator Layer, and Analytical Layer. The user uploads a PDF and submits a query to the Chatbot Based RAG Layer. This layer classifies the query. For standard queries, it returns a direct answer to the User Layer. For analytical queries, the Chatbot Based RAG Layer calls the Agent Layer, which extracts the relevant figures and structures them into a CSV. The CSV passes to the Validator Layer, which cross checks each figure against the source PDF. On failure, a notification returns to the User Layer. On success, the CSV moves to the Analytical Layer, which executes the query, generates a dashboard, and returns the result to the User Layer via the Chatbot Based RAG Layer.

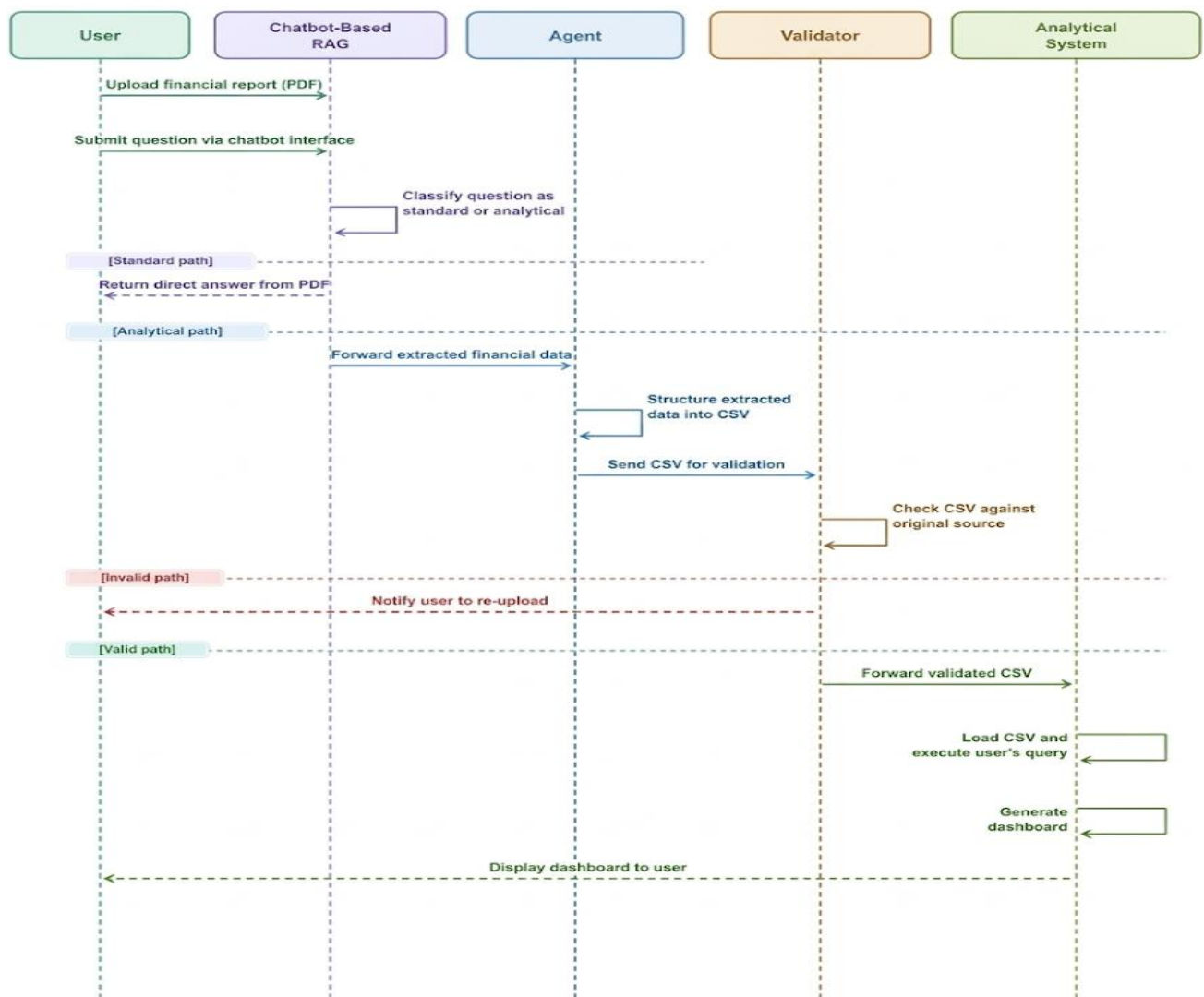


Figure 8. Sequence Diagram of Chatbot Based RAG for Analyzing Corporate Financial Data

Finally, a user activity diagram in figure 9 is provided to illustrate how the user interacts with the proposed artifact throughout the workflow. The process begins when the user uploads a financial report in PDF format and submits a question through the chatbot interface. The Chatbot Based RAG then classifies the query as standard or analytical. Standard queries return direct responses from the uploaded PDF. While, analytical queries, the system performs extraction, structuring, validation, and analytical processing before returning dashboards and analytical results to the user. Combined with the BPMN and sequence diagrams, the user activity diagram provides a clearer understanding of the user interaction flow within the proposed framework.

6.3.2.2 Structural Perspective



Figure 9. User Activity Diagram

The conceptual model presented in Figure 10 illustrates the architectural layers of the proposed framework and serves as the foundation for its implementation and evaluation. The process begins at the Input/Output layer, where the user submits a corporate annual report in PDF format alongside a natural language query. The uploaded PDF acts as the sole knowledge source, with no reliance on any external database, ERP system, or data warehouse.

The input is then passed to the Chatbot Based RAG layer, which classifies the query into one of three types. Out of scope queries notify the user that the requested data is not available in the document and terminate the process. Standard queries are answered directly from the PDF content and returned to the user without triggering further processing. Analytical queries activate the financial figure extraction component, which parses the PDF across three modalities, text segments, tables converted to structured CSV, and figures interpreted through a vision component, before forwarding the extracted data downstream.

The Agent layer then organizes the extracted financial figures into a structured and comparable CSV file, which is forwarded to the Validator layer. The Validator cross checks each extracted figure against its original location in the source PDF. If valid, the verified CSV is passed to the Analytical System. If invalid, the process is terminated and the user is prompted to re upload the document. The Analytical System loads the verified CSV, executes the user's query in a fact based manner without exposing the large language model to any raw PDF content, and generates a dashboard that is structured and comparable across reporting periods and companies. The dashboard is then returned to the user through the Input/Output layer with verified figures.

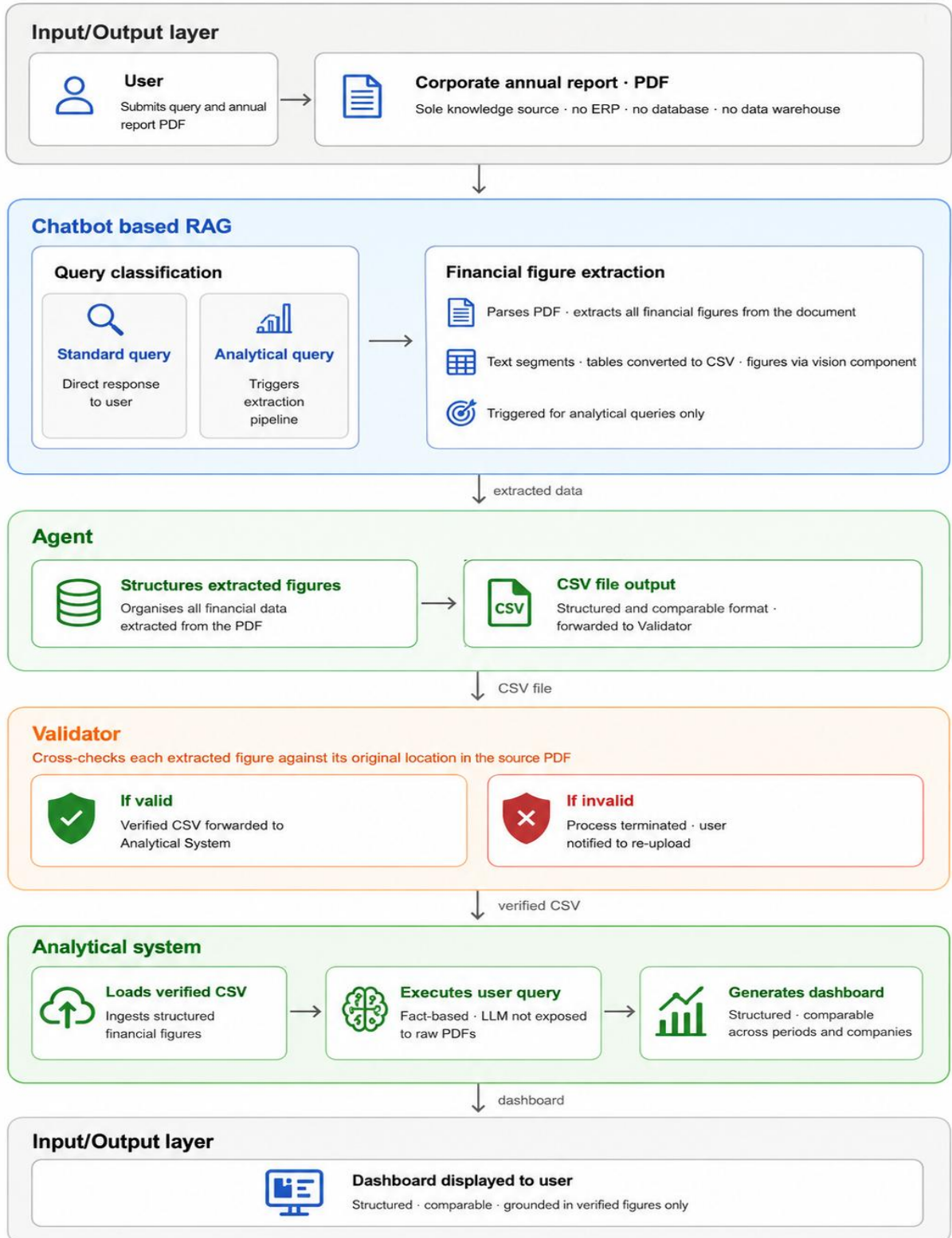


Figure 10. Conceptual Model of Chatbot Based RAG for Analyzing Corporate Financial Data

6.4 Evaluation

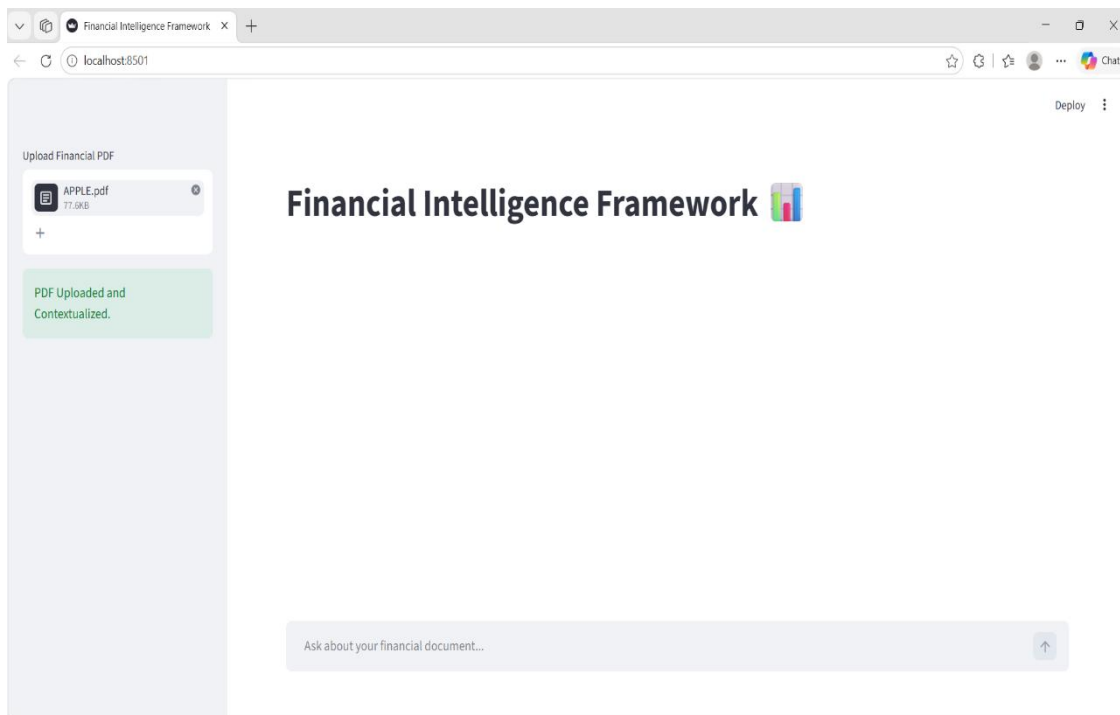
This chapter evaluates the proposed framework by assessing the conceptual model developed in [Chapter 6](#) against the research question and objectives defined in [Section 6.2](#). The evaluation is structured into two complementary components. The first is the Artifact Demonstration, which tests the practical feasibility of executing the proposed framework as a functioning system. The demonstration is divided into two parts: Validation, which examines whether the artifact fulfils the functional requirements defined in [Section 6.3.1.1](#) by verifying that each expected system behavior operates as intended, and in [Section 6.3.1.2](#) Verification, which provides a technical assessment of the artifact's runtime performance across indicators such as processing efficiency and error rate.

The second component is the Expert Interviews, in which domain practitioners are invited to evaluate both the conceptual model and the artifact demonstration, enabling iterative consultation with domain experts until agreement on the framework's utility is reached. The design validation cycle of Offermann et al. (2009) allows iteration back to the solution design phase if needed, while Hevner's (2004) rigor cycle ensures evaluation against established information systems criteria. The phase concludes with the summarisation and communication of results corresponding to Activity 6 of Peffers et al. (2007). Experts were selected following the Delphi Method outlined in [Section 5.1.2.1](#), ensuring participants were chosen based on the depth and relevance of their knowledge. Interview questions were structured according to the Technology Organization Environment (TOE) framework outlined in [Section 5.1.2.2](#) addressing the technological, organizational, and environmental dimensions of the proposed solution to capture a comprehensive and context sensitive assessment of its utility and applicability, with the full expert interview process and findings reported in [Section 6.4.2](#).

6.4.1 Artifact Demonstration

Step 1: User Initiated Document Submission

- This step corresponds to the first activity in the BPMN process flow in [section 6.3.2.1](#): “Upload PDF Financial Report”. It represents the process entry point. No subsequent activity within the Chatbot Based RAG is triggered until this step is successfully completed.
- Upon successful submission, the system ingests the document, extracts its full textual content, and retains it as the sole knowledge source for all downstream components within the pipeline. If no document is submitted, all subsequent process activities remain inactive.

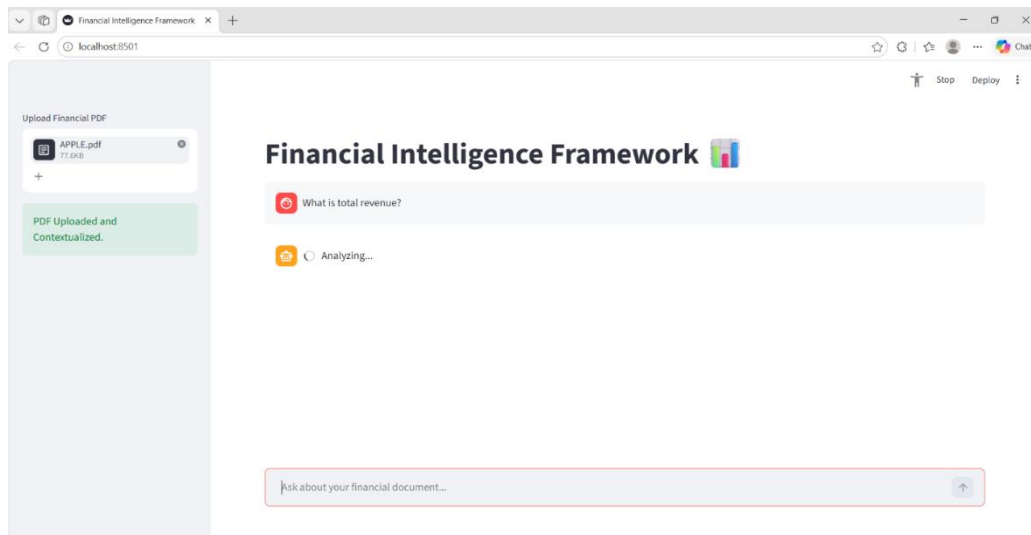


Code Snippet:

```
113 uploaded_file = st.sidebar.file_uploader("Upload Financial PDF", type="pdf")
114 pdf_context = ""
115
116 if uploaded_file:
117     with tempfile.NamedTemporaryFile(delete=False, suffix=".pdf") as tmp:
118         tmp.write(uploaded_file.read())
119         tmp_path = tmp.name
120
121     loader = PyPDFLoader(tmp_path)
122     pages = loader.load()
123     pdf_context = "\n".join([p.page_content for p in pages])
124     os.remove(tmp_path)
125     st.sidebar.success("PDF Uploaded and Contextualized.")
126
```

Step 2: Ask Question Via Chatbot Interface and Classify User Question

- Upon submission, the Chatbot Based RAG immediately classifies as shown in BPMN in [Section 6.3.2.1](#) the question by scanning it against a predefined set of analytical keywords. If a keyword match is detected, the question is classified as an Analytical query and routed accordingly. If no match is found, the question is classified as a Standard query and directed toward direct retrieval. This classification act as the exclusive gateway in the BPMN in [Section 6.3.2.1](#), determining which processing path the pipeline follows for all subsequent activities.



Code Snippet:

```
137 # User Input
138 if prompt := st.chat_input("Ask about your financial document..."):
139     # Save User Message
140     st.session_state.messages.append({"role": "user", "content": prompt})
141     with st.chat_message("user"):
142         st.markdown(prompt)
143
144     # Logic
145     with st.chat_message("assistant"):
146         if not pdf_context:
147             st.warning("Please upload a PDF first.")
148         else:
149             with st.spinner("Analyzing..."):
150                 classification = classify_question(prompt)
```

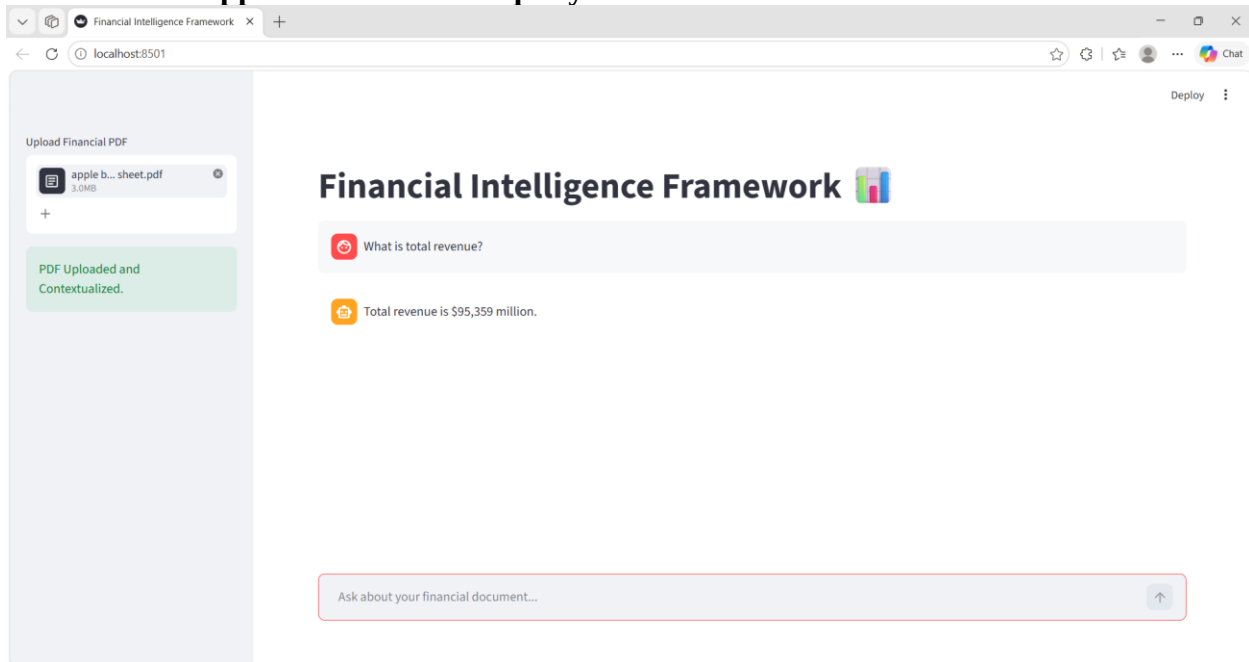
Step 3: Standard Query Direct Response from the Chatbot Based RAG

- Just like the BPMN in [Section 6.3.2.1](#) the Chatbot Based RAG processes the Standard query directly against the ingested PDF content and returns a response to the user. In case the requested information is not present, the user is notified within the conversational interface and the process terminates.

Code Snippet to the standard query to be found in and returned to the user:

```
152 # --- IF STANDARD ---
153 if classification == "STANDARD":
154     response = standard_agent(pdf_context, prompt)
155     st.markdown(response)
156     st.session_state.messages.append({"role": "assistant", "content": response})
157
158 # --- IF ANALYTICAL ---
159 else:
160     raw_extraction = extraction_agent(pdf_context, prompt)
161
162     if "[NOT_FOUND]" in raw_extraction:
163         msg = "Not in the PDF - it's not mentioned in the financial statement."
164         st.error(msg)
165         st.session_state.messages.append({"role": "assistant", "content": msg})
```

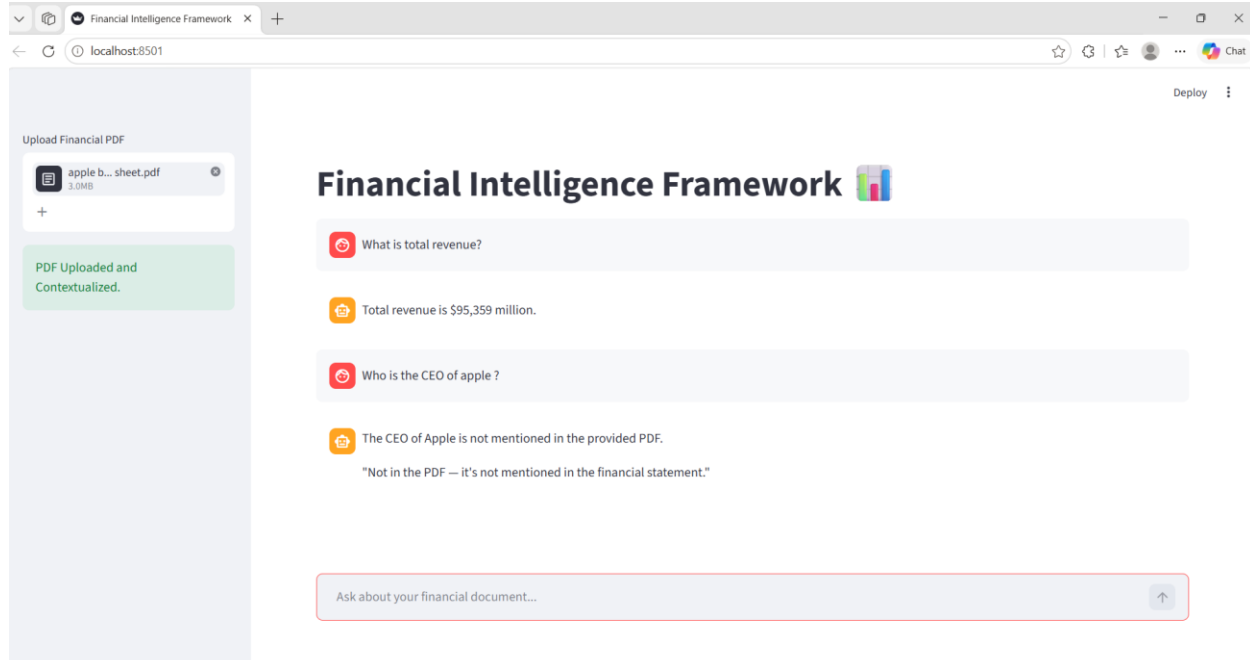
Framework Snippet to the standard query to be found in and returned to the user:



Code Snippet to the standard query to be out of scope or data not in the pdf:

```
36 def standard_agent(context, query):
37     """STANDARD QUESTION LOGIC: Direct RAG Retrieval."""
38     prompt = f"""
39     You are a financial assistant. Use the PDF context to answer.
40     CONTEXT: {context[:20000]}
41     QUESTION: {query}
42
43     RULE: If the answer is not in the PDF, return EXACTLY:
44     "Not in the PDF – it's not mentioned in the financial statement."
45     """
46     return llm.invoke(prompt)
```

Framework snippet to the standard query to be out of scope or data not in the pdf:



Step 4: Analytical Query Data Extraction and CSV Structuring by the Agent

- As you can refer to the BPMN in [Section 6.3.2.1](#) show the if statement that classifies the analytical path to the user. if the question is classified as Analytical, the Chatbot Based RAG extracts all relevant financial figures from the ingested PDF content and forwards them to the Agent. The Agent then structures the extracted figures into a CSV format, preparing the data for the Validator.

Code snippet for classifying query as an analytical query:

```
20 def classify_question(query):
21     """STRICT IF-STATEMENT BASED CLASSIFICATION using keywords."""
22     analytical_keywords = [
23         'compare', 'comparison', 'trend', 'analyze', 'analysis', 'calculate',
24         'percentage', 'growth', 'dashboard', 'graph', 'chart', 'visualize',
25         'ranking', 'rank', 'highest', 'lowest', 'average', 'summarize',
26         'grouped', 'aggregation', 'sql', 'query', 'across periods', 'over time',
27         'variance', 'difference', 'correlation', 'insights', 'forecast',
28         'breakdown', 'categories', 'segment', 'distribution', 'performance',
29         'top', 'bottom'
30     ]
31     query_lower = query.lower()
32     if any(keyword in query_lower for keyword in analytical_keywords):
33         return "ANALYTICAL"
34     return "STANDARD"
```

Code snippet for giving the Chatbot Based RAG extracting all relevant data and forwarding them to the agent to structure data in to a CSV:

```
48 def extraction_agent(context, query):
49     """ANALYTICAL EXTRACTION: Structures data into JSON."""
50     prompt = f"""
51     Context: {context[:25000]}
52     Task: Extract financial data for: {query}
53
54     RULES:
55     1. If data is missing, return EXACTLY: [NOT_FOUND]
56     2. Format Periods as: 'Month Day Year' (e.g., March 29 2025).
57     3. Use descriptive column names (e.g., 'Total Revenue', 'Net Sales').
58     4. Output ONLY a valid JSON list of objects.
59
60     EXAMPLE FORMAT:
61     [
62         [{"Period": "December 31 2024", "Net Income": 50000}],
63         [{"Period": "December 31 2023", "Net Income": 45000}]
64     ]
65     """
66     return llm.invoke(prompt)
```

Step 5: “Validation”

- Corresponding to the Validate CSV Check vs Source PDF activity in the Validator swimlane of the BPMN in [Section 6.3.2.1](#), the Validator cross checks every extracted figure in the CSV against the original uploaded PDF. If the data is valid, the verified CSV is forwarded to the Analytical System. If invalid, the user is notified within the conversational interface and prompted to re upload the document, after which the process terminates at one of the end events defined in the BPMN.

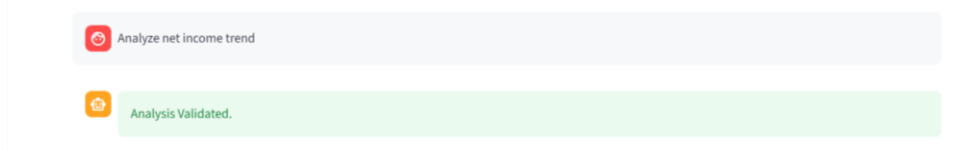
Code Snippet for defining a validation agent:

```
68 def validator_agent(context, df):
69     """VALIDATION AGENT: Compares extracted data against source."""
70     if df is None or df.empty:
71         return "FAIL"
72
73     data_summary = df.to_string()
74     prompt = f"""
75     Compare the following extracted data against the PDF context.
76     CONTEXT: {context[:10000]}
77     EXTRACTED_DATA: {data_summary}
78
79     TASK: Verify every number exists in the PDF.
80     If accurate, return 'PASS'. If any value is hallucinated or modified, return 'FAIL'.
81     """
82     response = llm.invoke(prompt).upper()
83     return "PASS" if "PASS" in response else "FAIL"
84
```

Code Snippet for Validation:

```
179         # VALIDATION STEP
180         validation_status = validator_agent(pdf_context, df)
181
182         if validation_status == "PASS":
183             excel_bin, csv_bin = dashboard_generator(df)
184             st.success("Analysis Validated.")
185             st.dataframe(df)
```

Framework snippet that data have been validated:



Code snippet in case if validation fails:

```
196         else:
197             msg = "Validation failed. Please reupload the document."
198             st.error(msg)
199             st.session_state.messages.append({"role": "assistant", "content": msg})
```

Step 6: Loading Validated Data into the Analytical System and Dashboard Generation

- once the Validator confirms the data as valid, the verified data is passed to the dashboard generator which produces an Excel file. In this implementation, the Excel file acts as both the data container and the Analytical System itself as shown in [Section 6.3.2.1](#) hosting the structured financial data and generating the dashboard output within the same environment. The structured data is displayed directly within the conversational interface as a table, and the user is provided with two downloadable outputs, the Excel Report containing the dashboard and the CSV file containing the raw structured data, concluding the process at the end event defined in the BPMN in [Section 6.3.2.1](#).

Framework Snippet:

The screenshot shows a web browser window with the address bar at localhost:8501. The page title is "Financial Intelligence Framework". On the left, there is a sidebar with an "Upload Financial PDF" section. A file named "apple b... sheet.pdf" (3.0MB) has been uploaded, and a green message states "PDF Uploaded and Contextualized." Below this, there is a "Deploy" button. The main content area displays a chat interface. The first message from the system (orange icon) states: "The CEO of Apple is not mentioned in the provided PDF. 'Not in the PDF — it's not mentioned in the financial statement.'" The second message from the user (red icon) is "Analyze net income trend". The third message from the system (orange icon) is "Analytical processing complete. Dashboard and files generated." Below this message is a table with two columns: "Period" and "Net Income". The table contains two rows of data. Below the table are two buttons: "Download Excel Report" and "Download CSV". At the bottom, there is a text input field with the placeholder "Ask about your financial document..." and a send button.

	Period	Net Income
0	March 29 2025	24780
1	March 30 2024	23636

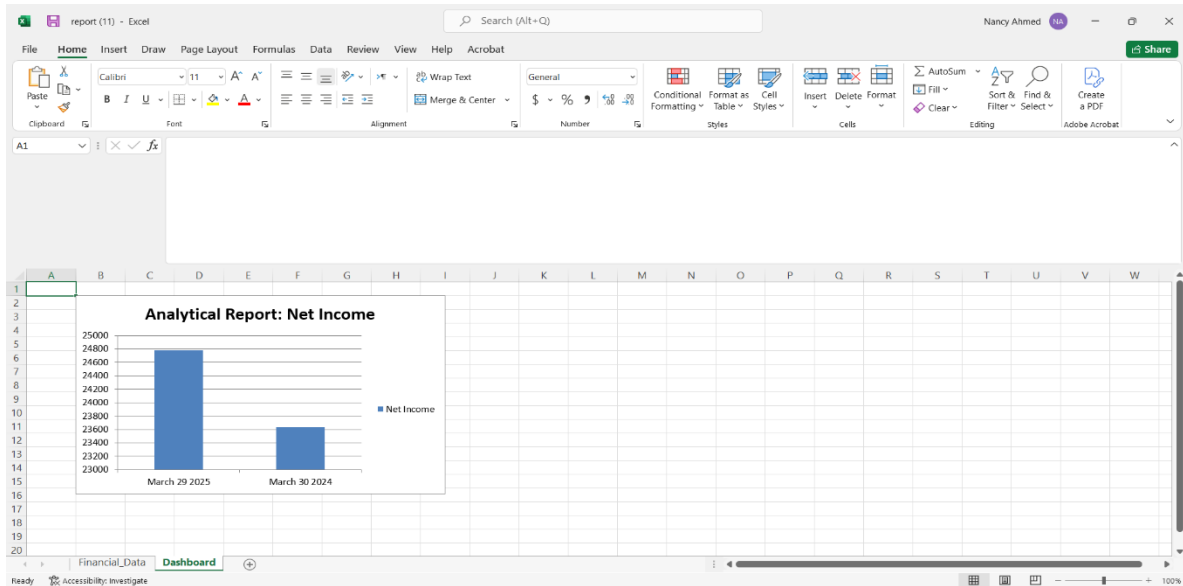
Extracted output 1: an Excel report containing 2 files. First, Financial data representing the structured financial data used to executed the user's query in a tabular that is used as the definitions for the dashboard form linking to the BPMN in [Section 6.3.2.1](#) the structured CSV.

The screenshot shows an Excel spreadsheet titled "report (11) - Excel". The spreadsheet has two columns: "Period" and "Net Income". The data is as follows:

Period	Net Income
March 29 2025	24780
March 30 2024	23636

The Excel interface includes the ribbon with tabs for File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Help, and Acrobat. The status bar at the bottom indicates "Ready" and "Accessibility: Investigate".

Extracted output 1: an Excel report containing 2 files. second, Dashboard representing the Dashboard of the structured financial data used to executed the user’s query linking to the BPMN in [Section 6.3.2.1](#)as the csv is sent to the analytical system however in this framework the csv and the analytical system is combined in the excel program for easier implementation.



Extracted output 2: an Excel report that contains data representing the structured financial data used to executed the user’s query in a tabular form linking to the BPMN in [Section 6.3.2.1](#) the query that the user may asked for also this is the output of the query if the user is wrote a SQL query.

Period	Net Income
March 29 2025	24780
March 30 2024	23636

6.4.1.1 Validation

The validation component examines whether the implemented system fulfils the functional requirements defined in [Section 6.3.1.1](#), assessed against the observed behavior of the artifact during the demonstration.

- **FR1 Multi Modal Document Parsing** was fulfilled. The system successfully parsed the uploaded PDF across text segments, financial tables converted to structured CSV, and figures processed through the vision language component, operating solely on the source document without any external infrastructure.
- **FR2 Natural Language Query Classification** was fulfilled. The Chatbot Based RAG correctly classified all submitted queries as standard, analytical, or out of scope, routing each through the appropriate processing path.
- **FR3 and FR9 Direct Response and Conversational Delivery** were fulfilled. Standard queries were answered directly from the ingested PDF and returned through the conversational interface, which served as the single unified interaction point for all outputs including responses, notifications, and downloadable files throughout the entire process.
- **FR4 and FR7 Agentic Extraction and Integrated Processing** were fulfilled. Upon detection of an analytical query, the agent autonomously extracted the relevant financial figures and structured them into a CSV file, treating the annual report as a unified hybrid source and simultaneously handling both structured numerical figures and unstructured textual narratives.
- **FR5 and FR8 Binary Validation and Hallucination Containment** were fulfilled. The validator successfully caught and rejected incorrectly extracted figures during the demonstration, preventing them from reaching the analytical system. The LLM operated exclusively over validated data at all times, never exposed to raw PDF content during numerical output generation.
- **FR6 Automated Dashboard and Structured Output Generation** was fulfilled. Validated data was loaded into an Excel file, a dashboard was automatically generated in response to the user's query, and both the Excel report and raw CSV were returned as downloadable outputs through the chat interface.

- **FR10 On Premise Deployment** was fulfilled. No document content, extracted figure, or query data was transmitted to any external cloud service or third party API at any stage, confirming full compliance with data sovereignty requirements.

All functional requirements were successfully demonstrated, confirming that the artifact behaves as specified in the conceptual design.

6.4.1.2 Verification

The verification provides a qualitative technical assessment of the artifact's runtime performance based on structured observation during the demonstration session, organized across four dimensions. Regarding processing and response time, in earlier development iterations each pipeline stage produced visible intermediate screens for upload, Analyzing, validation, and extraction, which introduced noticeable latency across the end to end process, however by the final version these transitions were largely eliminated, resulting in a substantially smoother user experience in which processing occurred without interrupting the conversational flow.

Regarding validation and extraction accuracy, the validator operated correctly as a binary gatekeeper throughout the demonstration, where incorrectly extracted figures were successfully caught and rejected, preventing inaccurate data from reaching the analytical system and confirming that the hallucination mitigation mechanism functions as intended and that the pipeline enforces the requirement that the LLM operates only over verified structured data.

Regarding error handling, the system demonstrated appropriate behavior at each defined failure point, where out of scope notifications, validation rejection pathways, and re upload prompts were all structurally integrated and functioned as expected, with no unhandled errors or pipeline interruptions encountered during the demonstration session. Finally, regarding overall performance, the artifact was assessed at a performance level of 7 out of 10, reflecting reliable and consistent execution across all core pipeline stages including document ingestion, query classification, extraction, validation, and output generation, with the score acknowledging remaining scope for optimization particularly in processing speed under larger or more complex documents and in extraction robustness across a wider variety of annual report formats.

6.4.2 Experts Interviews

To obtain a more validated assessment of the proposed framework, expert interviews were conducted to qualitatively evaluate the practicality, relevance, and expected value of the RAG based financial data analysis system, enabling iterative consultation with domain experts until agreement on the framework's utility was reached. The design validation cycle of Offermann et al. (2009) allows iteration back to the solution design phase if needed, while Hevner's (2004) rigor cycle ensures evaluation against established information systems criteria. The phase concludes with the summarisation and communication of results corresponding to Activity 6 of Peffers et al. (2007).

The interviews were conducted with five candidates drawn from different organizational and technical roles, ensuring that the evaluation captured both the system's technical soundness and its practical organizational applicability. Prior to the interviews, technical participants were provided with the LLM architecture, the behavioral model, and the conceptual model outlined in Chapter 6, giving them a clear understanding of the system's design and interaction flow. All participants were presented with a live demonstration of the artifact, allowing them to observe the pipeline in operation before sharing their assessment. This ensured that all feedback was grounded in direct exposure to both the theoretical design and its implemented form.

Experts were selected following the Delphi Method as described in Section 5.1.2.1, which requires that participants be chosen based on the depth and direct relevance of their expertise rather than through convenience sampling alone. The selected panel comprised the following profiles:

- A Marketing Manager at Bayer Limited Egypt, representing an end user organizational perspective.
- An Accounting Teaching Assistant at the German University in Cairo, representing academic and financial domain knowledge.
- A Digital Transformation Teaching Assistant at the German University in Cairo, representing expertise in organizational technology adoption.

- A Technical Manager at Capgemini, representing enterprise level technical and systems integration expertise.
- A Fintech Manager at Etisalat, representing applied financial technology and industry deployment experience.

Interview questions were structured according to the Technology, Organization, and Environment (TOE) framework as described in [Section 5.1.2.2](#), guiding both question design and response analysis. Technical participants were directed toward the technological dimension, organizational participants interacted directly with the artifact demonstration and addressed the organizational dimension, and all participants were asked the same questions under the environmental dimension given their shared institutional and industry context. Questions were kept broad to allow participants to respond freely and surface considerations beyond those anticipated by the researcher.

✦ **Technology**

Questions in this dimension explored the overall technical soundness of the proposed architecture, the appropriateness of the RAG approach for financial document processing, the effectiveness of the validation mechanism as a hallucination mitigation strategy, the suitability of the on premise deployment model, and whether the defined functional requirements adequately captured the system's intended capabilities.

All participants confirmed the relevance and technical soundness of the proposed framework, highlighting that it directly addresses a recognized gap in how organizations currently process corporate financial documents. The separation between the retrieval pipeline and the LLM reasoning layer was received positively, with participants noting that it reflected a sound understanding of where hallucination risk originates in language model systems. The validation mechanism was considered as an implementable safeguard, and participants broadly agreed that it tackled one of the most critical limitations of LLM based financial applications in a traceable and auditable way.

On the functional requirements, participants agreed that the defined requirements were comprehensive and well aligned with the system's objectives. A recurring suggestion was that future iterations could extend the validation mechanism to incorporate confidence scoring, which would provide better insights into extraction quality across varying document formats and structures. The on premise deployment model was acknowledged as technically demanding but well justified given the sensitivity of financial data, and participants noted that it meaningfully differentiates the framework from cloud dependent alternatives. Overall, the framework was considered innovative and directly applicable to real world financial analysis contexts.

✦ **Organization**

Questions in this dimension addressed the usefulness of the system for day to day financial analysis tasks, the accessibility of the conversational interface for non technical users, the degree to which the system would reduce manual reporting effort, the reliability of the Excel dashboard as a decision support tool, and whether the framework as demonstrated would be considered suitable for deployment within their respective organizations.

All participants emphasized the practical value the system would bring to organizational workflows, particularly in reducing the time and effort currently associated with manual extraction and report preparation from annual reports. The ability to submit a natural language query and receive a structured, validated, and downloadable Excel output was consistently identified as a meaningful improvement over existing processes, and participants noted that it would allow employees to redirect effort toward higher value analytical tasks rather than data gathering. The conversational interface was well received across the group, with participants highlighting its accessibility as a key enabler for non technical staff who regularly work with financial documents but lack a technical background.

There was some variation in perspective regarding deployment readiness. While participants broadly agreed that the framework demonstrated strong foundational capability, some felt that additional testing across a more diverse range of document formats and organizational data environments would be necessary before recommending full scale adoption. The Excel dashboard output was viewed as practically well aligned with existing organizational workflows, as it delivered results in a format already familiar to finance and management teams, minimizing the adaptation required on the user side.

✦ *Environment*

All five participants were asked the same set of questions in this dimension, reflecting the shared institutional and industry context within which both technical and organizational practitioners operate. Questions addressed the broader regulatory and competitive environment surrounding AI adoption in financial analysis, the influence of data privacy requirements on the viability of cloud based versus on premise solutions, the current state of AI literacy within target organizations, and whether prevailing industry trends were likely to accelerate or hinder adoption of systems of this kind.

Participants across both profiles reached a strong degree of agreement that data privacy and sensitivity represent the primary environmental driver favoring on premise deployment, and that organizations handling corporate financial data face increasing inspection of where and how that data is processed. The on premise architecture was therefore considered environmentally well positioned, particularly in regulated industries and in organizations operating under strict internal data governance policies.

Participants also broadly agreed that AI proficiency within organizations remains uneven, and that successful adoption would depend not only on technical performance but on change management, user training, and organizational trust in AI generated outputs. A recurring concern across the group was the risk of over reliance on system outputs, with several participants noting that users could become dependent on the framework in a way that diminishes their own analytical engagement. It was suggested that this be addressed through transparent system design and appropriate user guidance.

The current industry trend toward AI assisted decision making was acknowledged as creating a favorable adoption environment, and the conversational interface was once again highlighted as a meaningful enabler in this context given that it reduces the technical overhead required for interaction. There was no significant points of disagreement emerged across the environmental dimension, reflecting a shared understanding of the external conditions shaping the framework's deployment context.

7 Limitation and Future Recommendations

Every research artifact operates within boundaries that define the scope of its claims and the conditions under which its findings hold. This chapter acknowledges the limitations inherent to the proposed framework as developed within the constraints of this research, and presents the future recommendations that follow from them. [Section 7.1](#) addresses the limitations related to the concept and scope of the framework, and [Section 7.2](#) presents the future recommendations.

7.1 Limitations related to the concept

The proposed framework carries several limitations that are important to acknowledge in the interest of accurately representing the scope of this research. The artifact was developed and evaluated at a small prototype scale, meaning that the demonstration does not cover the full range of functional requirements under real world operating conditions, and the system's behavior under larger document volumes, more complex queries, or sustained use remains untested. Moreover, the framework operates without connection to any ERP system or persistent database, meaning that all session data is held in memory and lost once the session ends. No extracted figures, prior queries, or generated outputs are retained between interactions, which limits the system's continuity in organizational contexts where data persistence is expected.

From a functional standpoint, not all possible user query scenarios were anticipated in the current design. Queries that fall outside the defined classification boundaries or that combine retrieval and extraction in unexpected ways may expose the system to residual hallucination risk that the current pipeline does not fully eliminate. The visualization capability is similarly constrained, as the current implementation generates only bar graph outputs within the Excel file and does not yet support multiple dashboard types or varied chart formats, requiring further development before the output layer can be considered comprehensive. The validation mechanism also operates solely as a binary pass or fail check without producing any quantitative confidence scoring, leaving users without a measure of the degree of certainty associated with a validated extraction.

Finally, the system accepts a limited number of PDF pages per session due to current memory constraints, restricting its applicability to shorter documents and preventing it from processing the full length of many corporate annual reports without modification. Input and output format support is also narrow, as the system currently accepts documents exclusively in PDF format and delivers structured outputs exclusively in Excel, limiting flexibility for users who operate within different document and reporting ecosystems.

7.2 Future Recommendations

Several directions for future development emerge directly from the limitations identified in [Section 7.1](#). The most immediate priority is the integration of a persistent database layer, such as Supabase, to store extracted figures, validated outputs, and generated reports in real time as the process runs rather than losing all data when the session ends. This would also lay the groundwork for full process automation, enabling the pipeline to automatically schedule document ingestion, execute extraction and validation cycles, and deliver outputs without requiring manual re initiation at each stage.

Beyond infrastructure, the validation mechanism should be extended to incorporate confidence scoring rather than a binary result alone, and the dashboard generation layer should be expanded to support varied visualization types beyond bar graphs. Broadening accepted input formats beyond PDF and output formats beyond Excel would further increase flexibility across different organizational environments. ERP integration and more comprehensive query classification coverage are also recommended as longer term development priorities to reduce residual hallucination risk and contextualize extracted data against live organizational systems.

8 Conclusion

This research set out to address a clearly identified scientific gap: the absence of a system capable of extracting structured financial data embedded within unstructured corporate annual reports and presenting it in a structured and comparable output format through a conversational AI interface, deployed entirely on premise, with active hallucination mitigation in place.

To address this gap, a framework was designed using retrieval augmented generation and large language models, structured across five layers: an input/output layer, a chatbot based RAG layer, an agent layer, a validator layer, and an analytical system. The framework treats corporate annual reports as a unified hybrid source, simultaneously handling both structured numerical figures and unstructured textual narratives, with the LLM operating exclusively over verified structured data at all times to structurally contain the risk of hallucinations in numerical outputs.

The artifact demonstration confirmed that all ten functional requirements were successfully fulfilled, and the verification assessment rated the artifact at a performance level of 7 out of 10, reflecting reliable execution across all core pipeline stages while acknowledging remaining scope for optimization in processing speed and extraction robustness. Expert interviews conducted with five domain practitioners structured according to the Technology, Organization, and Environment framework confirmed the technical soundness of the architecture, the practical value of the conversational interface for non technical users, and the importance of the on premise deployment model in regulated corporate environments.

The comparative analysis of Kong et al. (2024), Duane et al. (2025), Al et al. (2025), Yang (2022), and Srinivasan et al. (2025) confirmed that persistent gaps in hallucination mitigation, structured data extraction from mixed format documents, and on premise deployability remain unresolved across the existing literature, and that this research directly addresses those gaps through a practical framework design confirmed through artifact demonstration and expert evaluation.

Several limitations were identified, including the small prototype scale of the artifact, the absence of persistent data storage across sessions, visualization support limited to bar graph outputs within the Excel file, and document support confined to PDF format with structured outputs delivered exclusively in Excel. These limitations define the future development directions outlined in [Chapter 7](#).

Overall, this research demonstrates that a chatbot based RAG framework is a viable and practically relevant approach to extracting structured financial data from unstructured corporate annual reports within a fully on premise environment, offering a foundation upon which future development can build toward a fully deployable corporate solution.

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Appendix A

The following appendix documents the full set of interview questions posed to each participant, organized according to the Technology, Organization, and Environment (TOE) framework. Questions were kept intentionally broad to allow participants to respond freely. Participants are identified by their professional role rather than by name to maintain consistency with the anonymized reporting approach adopted in Section 6.4.2.

✦ *Participant Profiles*

Participant Role		Organization Type
P1	Marketing Manager	Corporate Organization
P2	Accounting Teaching Assistant	German University in Cairo
P3	Digital Transformation Teaching Assistant	German University in Cairo
P4	Technical Manager	Capgemini
P5	Fintech Manager	Etisalat

✦ *Technological Dimension*

(Addressed to P3, P4, and P5)

Q1. After reviewing the conceptual model and behavioral model, how would you assess the overall technical soundness of the proposed framework?

P3: The framework is well thought out. The decision to keep the LLM separate from raw numerical data is particularly strong and shows awareness of where language models tend to fail in financial contexts.

P4: From an enterprise systems perspective, the architecture is coherent. The layered design is clean and each component has a clearly defined responsibility, which is important for maintainability and future scaling.

P5: The RAG approach is the right choice for this use case. Financial documents are dense and highly structured in parts, and a retrieval augmented approach is far more appropriate than relying on a model's parametric memory alone.

Q2. How effective do you consider the binary validation mechanism as a strategy for mitigating hallucinations in numerical outputs?

P3: It is effective within its current scope. The idea of cross checking every extracted figure against the source before it reaches the model is sound. My only thought is that a confidence score alongside the binary result could add more depth to the output.

P4: It is a pragmatic solution. It may not catch every edge case, particularly in documents with unusual formatting, but as a first line of defense it is well designed and traceable.

P5: In fintech, auditability is everything. The fact that every figure can be traced back to a validated source is a significant advantage and addresses a concern that comes up constantly when organizations consider deploying AI for financial reporting.

Q3. Do you consider the functional requirements as defined to be comprehensive and well aligned with the system's objectives? Are there any gaps you would identify?

P3: They are comprehensive. I would consider adding a requirement around multi document querying in future versions, as comparative analysis across multiple annual reports would significantly extend the system's utility.

P4: Well aligned overall. The on premise requirement is particularly important and I would not treat it as optional given the sensitivity of the data being processed.

P5: The requirements are solid. The main gap I see is around confidence scoring in the extraction layer, which would make the validation output more informative for technical users reviewing the pipeline.

Q4. How do you assess the scalability and feasibility of the on premise deployment model for organizations of varying sizes?

P3: Feasible for medium to large organizations with dedicated IT infrastructure. Smaller organizations may struggle with the hardware requirements, which is worth noting as a limitation.

P4: Scalability is achievable but requires upfront investment. For enterprise clients this is not unusual, and the data sovereignty argument more than justifies the infrastructure cost.

P5: In regulated financial environments, on premise is often not optional. The framework being designed around this from the outset rather than as an afterthought is the right approach.

✦ **Organizational Dimension**

(Addressed to P1, P2, and P3)

Participants in this section interacted directly with the artifact demonstration before responding.

Q1. Based on the demonstration, how useful do you consider this system for day to day financial analysis tasks within an organization?

P1: Very useful. As someone who regularly works with reports but does not have a technical background, being able to ask a question in plain language and get a structured output is genuinely valuable. It removes a significant bottleneck.

P2: From an accounting perspective, the extraction of financial figures into a comparable spreadsheet format is extremely practical. A lot of time is currently spent doing this manually and this system addresses that directly.

P3: It is useful at both the operational and strategic level. Operationally it saves time. Strategically it enables faster access to financial insights which supports better decision making.

Q2. How accessible do you consider the conversational interface for non technical users within an organization?

P1: Very accessible. The interface requires no technical knowledge and the outputs are in a format that anyone working with financial data would immediately recognize.

P2: It is straightforward. The fact that it returns an Excel file means there is essentially no learning curve for the output side. The input side is just typing a question, which is intuitive.

P3: The accessibility is one of its strongest points. Lowering the barrier to AI assisted analysis is important for organizational adoption, and this interface does that effectively.

Q3. To what extent do you think this system would reduce manual effort in financial reporting and data extraction within your organization?

P1: Substantially. The process of extracting figures from annual reports and preparing them for presentation currently takes considerable time. This system would compress that significantly.

P2: The reduction in manual effort would be meaningful, particularly during reporting periods when the volume of documents being processed is high.

P3: It would free up analyst time considerably. The value is not just in speed but in consistency, since manual extraction introduces human error that this system eliminates through validation.

Q4. Would you consider the framework as demonstrated to be ready for deployment within your organization, and what conditions would need to be met for adoption?

P1: Not immediately, but close. I would want to see it tested on a wider range of document types before recommending it formally. The foundation is strong though.

P2: With some further testing and documentation for end users, yes. The core functionality is there.

P3: Deployment readiness is high from a functional standpoint. The main condition for adoption in my context would be formal validation across a larger document set and a structured onboarding process for users.

✦ **Environmental Dimension**

(Addressed to all participants: P1, P2, P3, P4, P5)

Q1. How do you assess the influence of data privacy concerns and regulatory requirements on the viability of AI based financial analysis tools in your industry or context?

P1: Privacy is a primary concern in any corporate environment. The fact that this system does not send data outside the organization is the single most important factor in making it viable for us.

P2: Academic institutions handle sensitive data and there is increasing awareness of where AI tools are sending information. An on premise system removes that concern entirely.

P3: Regulatory pressure around data handling is only increasing. Systems that are designed with privacy as a core constraint rather than an add on are going to be far more viable going forward.

P4: In enterprise consulting, data governance is non negotiable for financial clients. The on premise architecture is not just a technical choice here, it is a compliance enabler.

P5: In fintech, regulatory scrutiny of AI tools is significant and growing. The ability to demonstrate that no financial data leaves the organization's infrastructure is a material advantage in that environment.

Q2. How would you characterize the current level of AI literacy within organizations of the type that would use this system, and how does that affect adoption potential?

P1: AI literacy is improving but still uneven. Most people understand that AI tools exist but fewer understand how to evaluate their outputs critically. That gap needs to be addressed through training.

P2: At the university level there is growing awareness, but in traditional finance departments the familiarity with AI is still limited. Adoption would require deliberate change management.

P3: Digital transformation initiatives are raising AI literacy gradually but the pace varies significantly by organization and industry. A system with a simple interface like this one is better positioned for adoption than one that requires technical knowledge to operate.

P4: Enterprise organizations have invested heavily in AI literacy programs over the past few years, but the gap between awareness and confident use remains wide. User guidance and transparent system design are essential.

P5: In fintech the literacy level is higher than average, but even there, trust in AI generated financial outputs has to be earned through demonstrated accuracy and auditability.

Q3. Do you think prevailing industry trends and competitive pressures are likely to accelerate or slow the adoption of AI assisted financial analysis tools?

P1: Accelerate. Organizations that are not exploring AI for financial analysis are already falling behind, and that pressure is only going to increase.

P2: Accelerate, particularly as more tools become accessible and the outcomes become more visible. Demonstrated value drives adoption faster than any other factor.

P3: Accelerate significantly. Digital transformation is no longer optional in most industries and financial analysis is one of the areas where AI can deliver immediate, measurable value.

P4: Accelerate. The competitive dynamic in consulting is already shifting toward AI augmented delivery and clients are beginning to expect it as a baseline capability.

P5: Accelerate, but with caution. The fintech sector is enthusiastic about AI adoption but the regulatory environment requires that adoption be deliberate and well documented.

Q4. What risks or challenges do you foresee in the broader adoption of a system of this kind, beyond the technical and organizational factors already discussed?

P1: The main risk is over reliance. If users stop critically reviewing outputs because they trust the system, errors that do occur will be harder to catch.

P2: Trust is the central challenge. Financial outputs carry significant consequences and users need to be confident in the system before they act on its results.

P3: Change resistance is a real factor. People with established workflows are often reluctant to adopt new tools even when the benefit is clear, and that requires organizational investment in transition management.

P4: Integration with existing enterprise systems is a practical challenge that should not be underestimated. The system works well as a standalone tool but its value increases significantly when it can interact with broader data infrastructure.

P5: Regulatory uncertainty around AI in financial reporting is a risk. The frameworks governing what AI tools can and cannot do in this space are still evolving, and organizations need to be prepared to adapt.